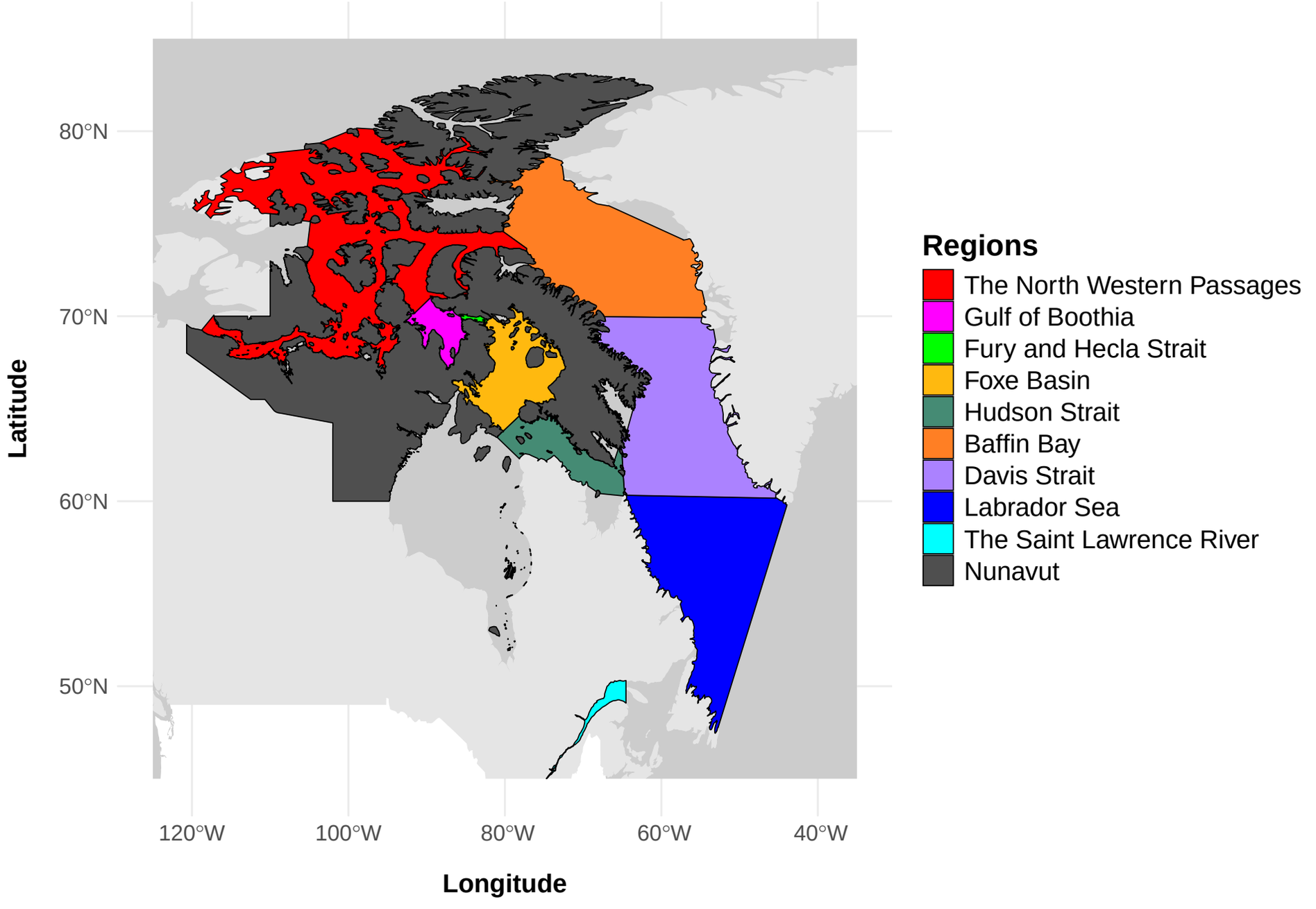
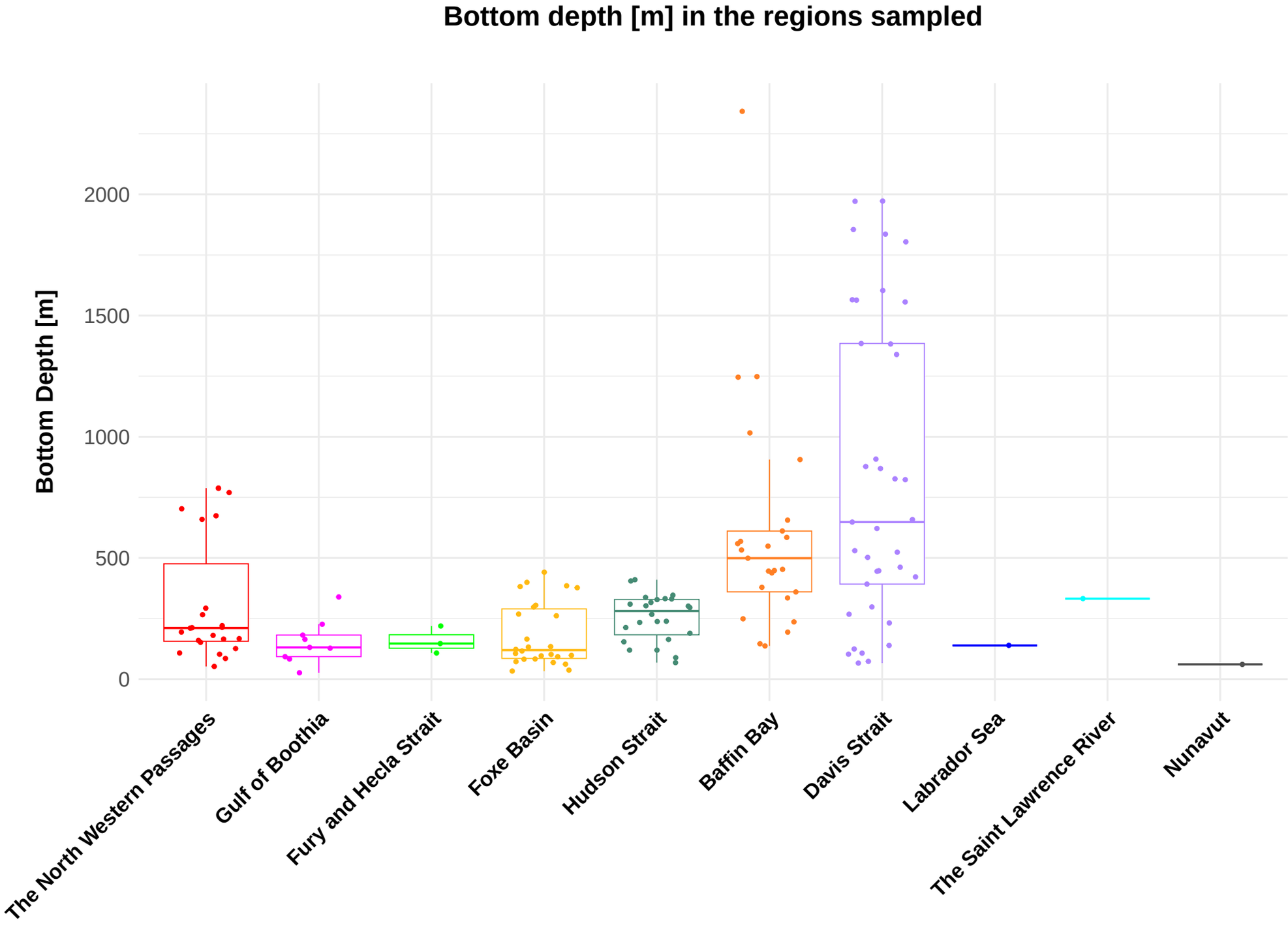
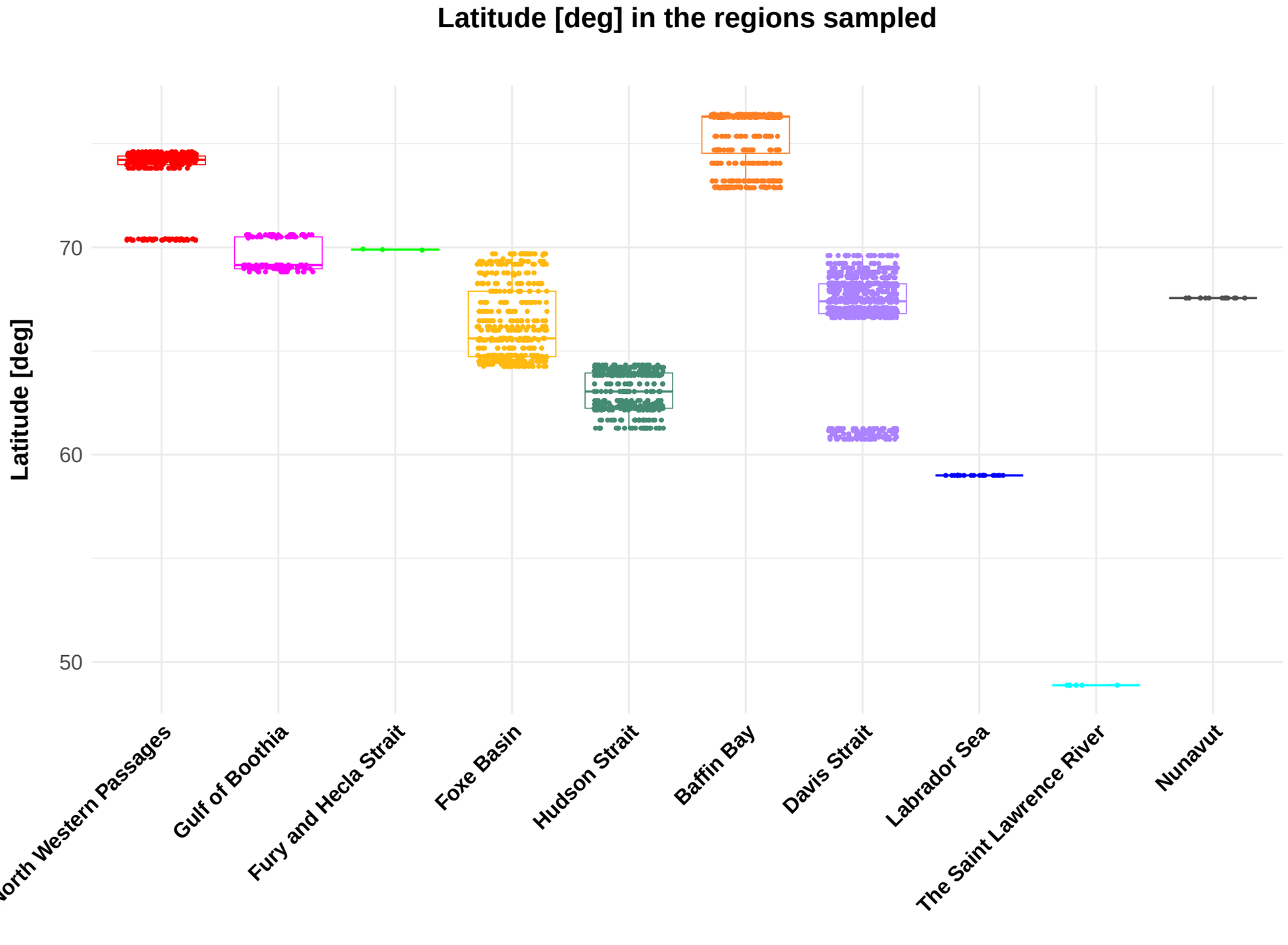
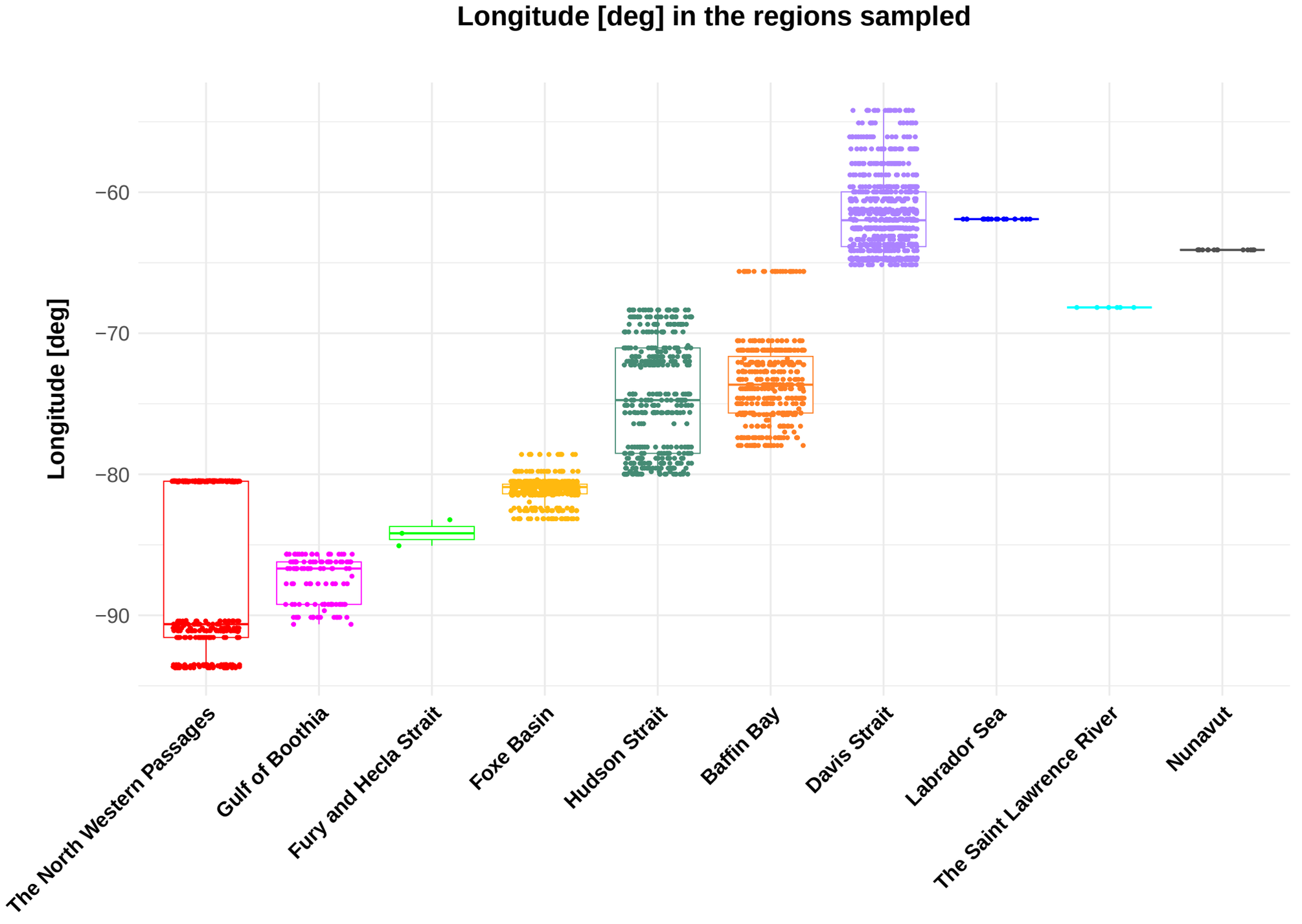
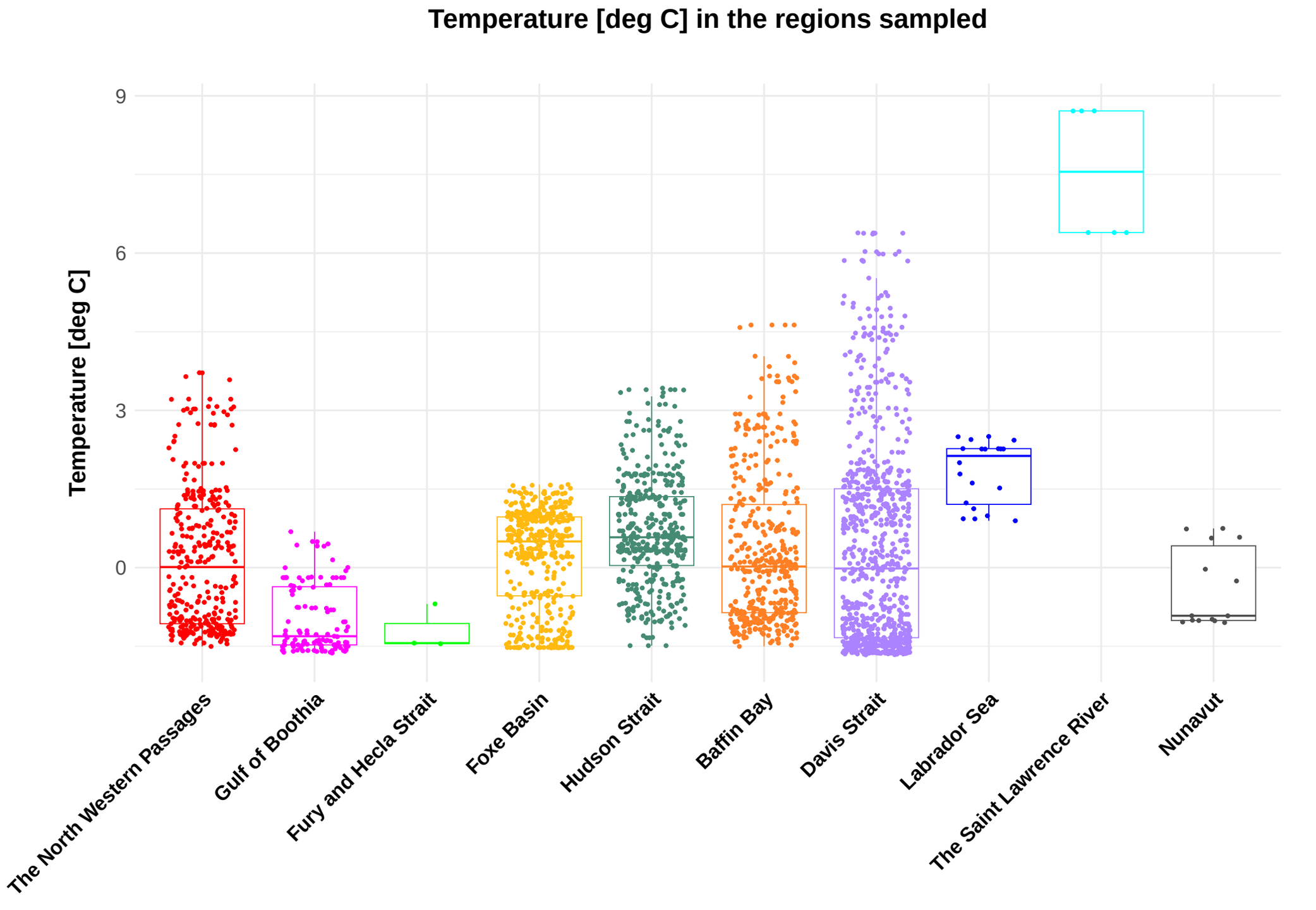
Regions sampled during Leg 2 and 4 of Amundsen 2023

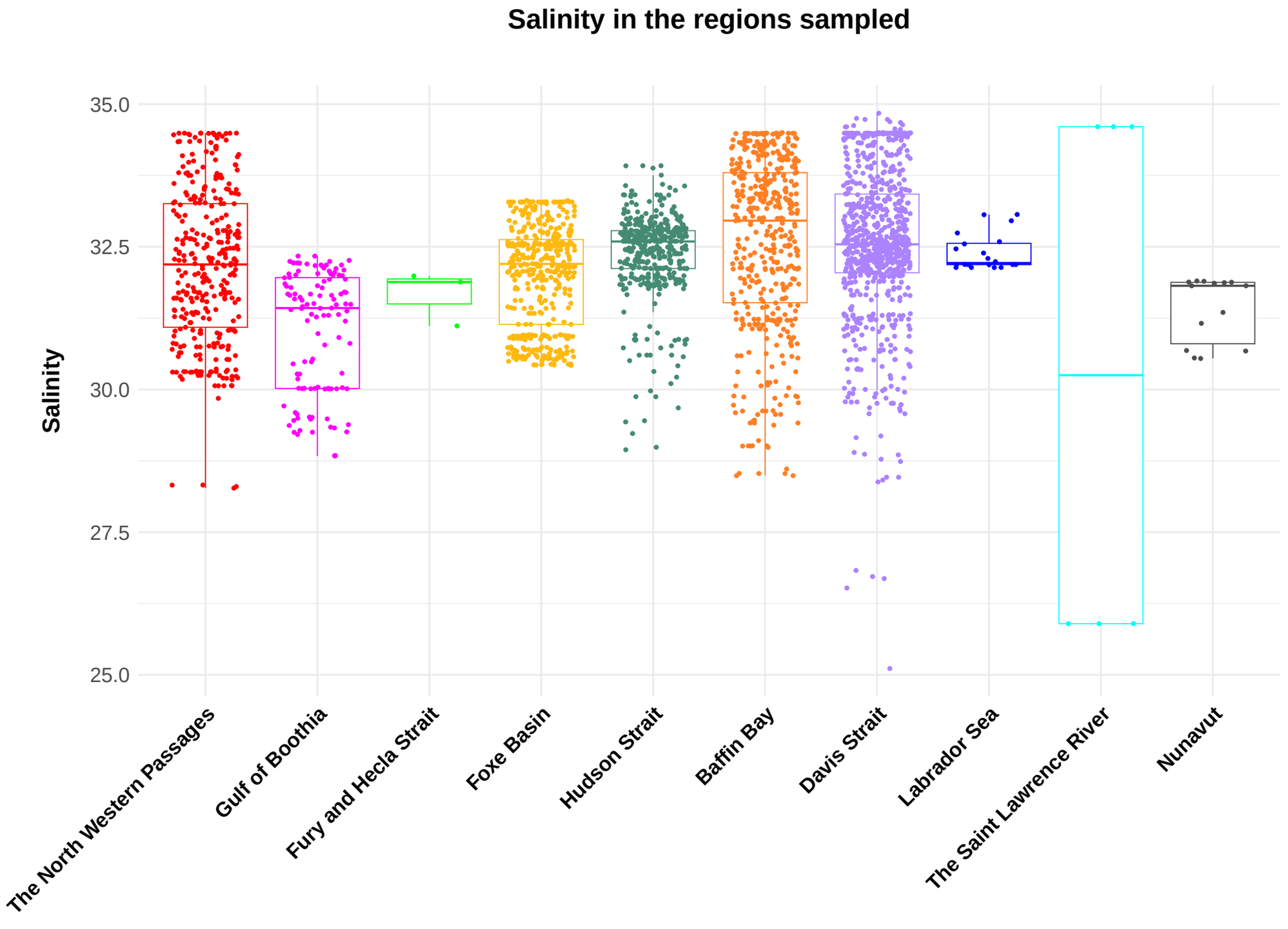


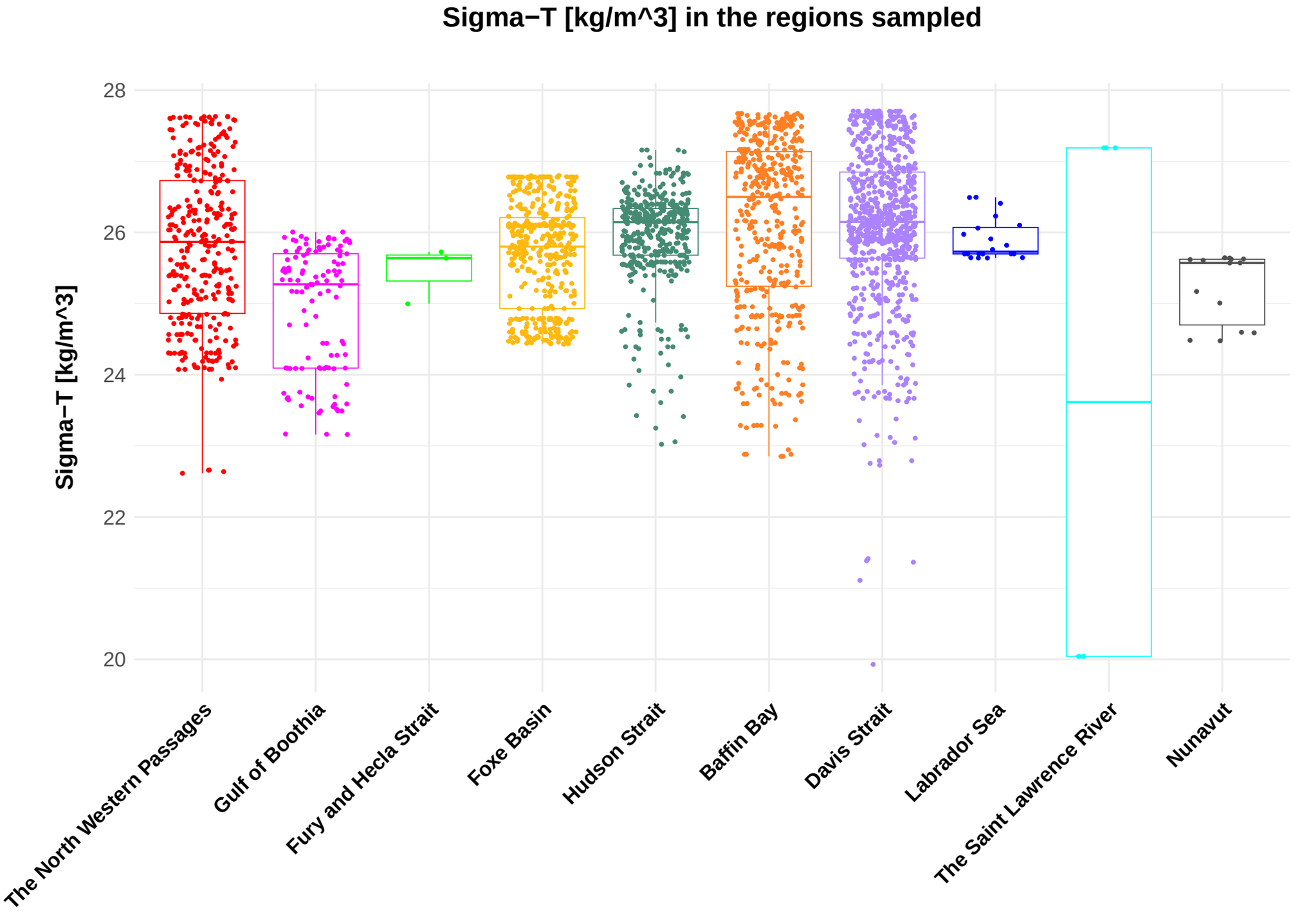

Bottom depth [m] in the regions sampled
Bottom Depth [m]
2000
1500
1000
500
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

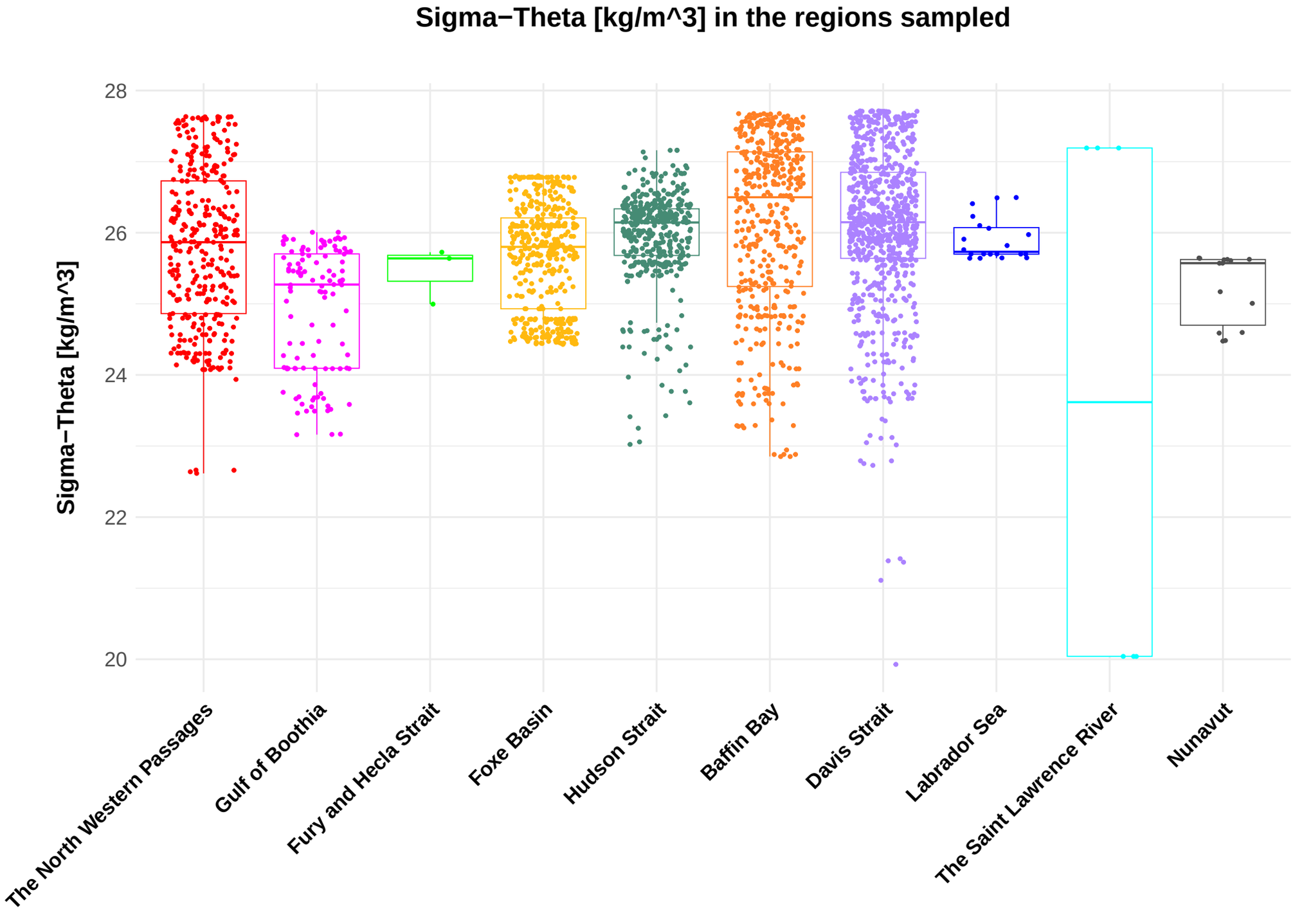

Latitude [deg] in the regions sampled
Latitude [deg]
70
60
50
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

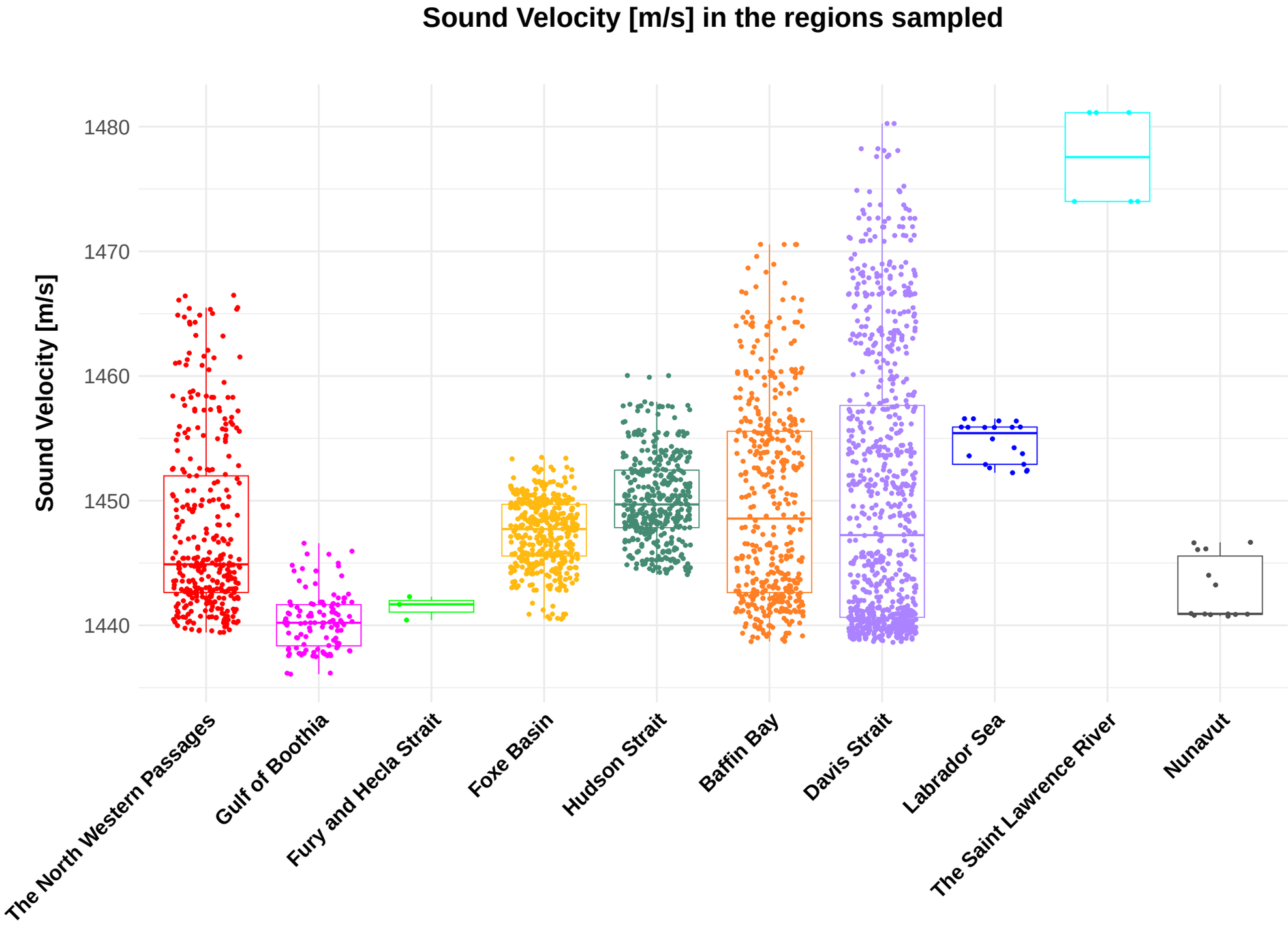

Longitude [deg] in the regions sampled
Longitude [deg]
−60
−70
−80
−90
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

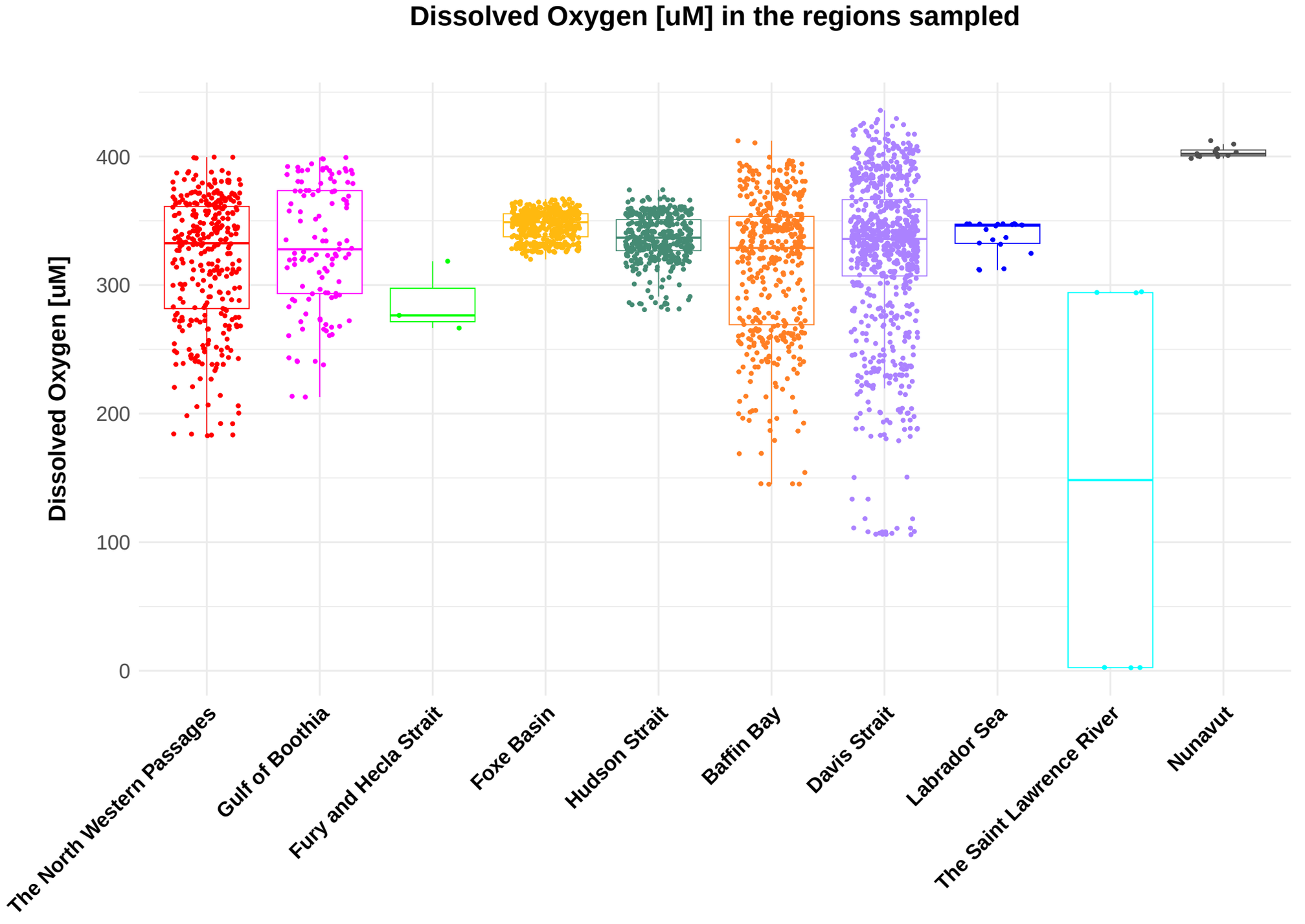

Temperature [deg C] in the regions sampled
Temperature [deg C]
9
6
3
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

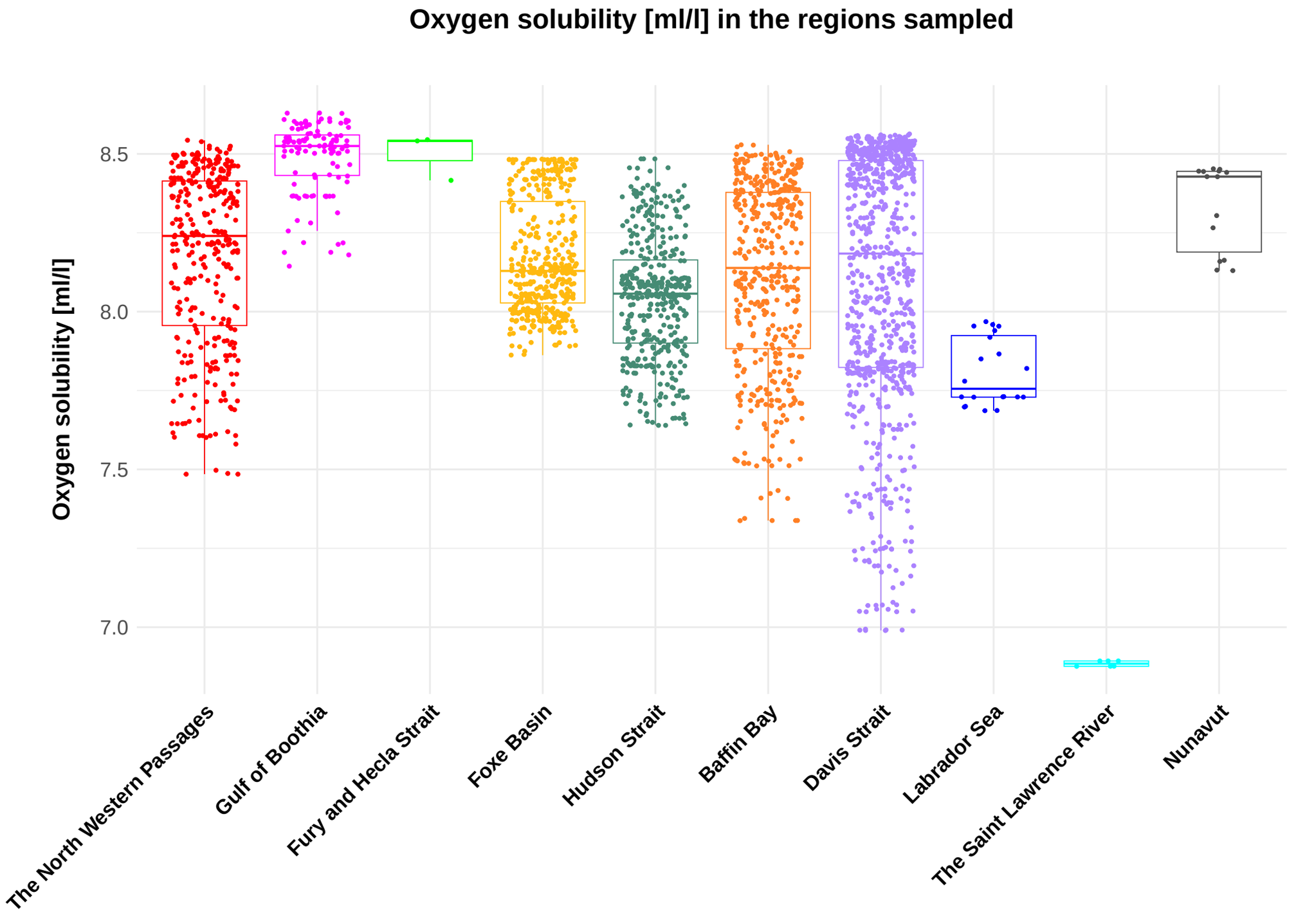

Salinity in the regions sampled
Salinity
35.0
32.5
30.0
27.5
25.0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

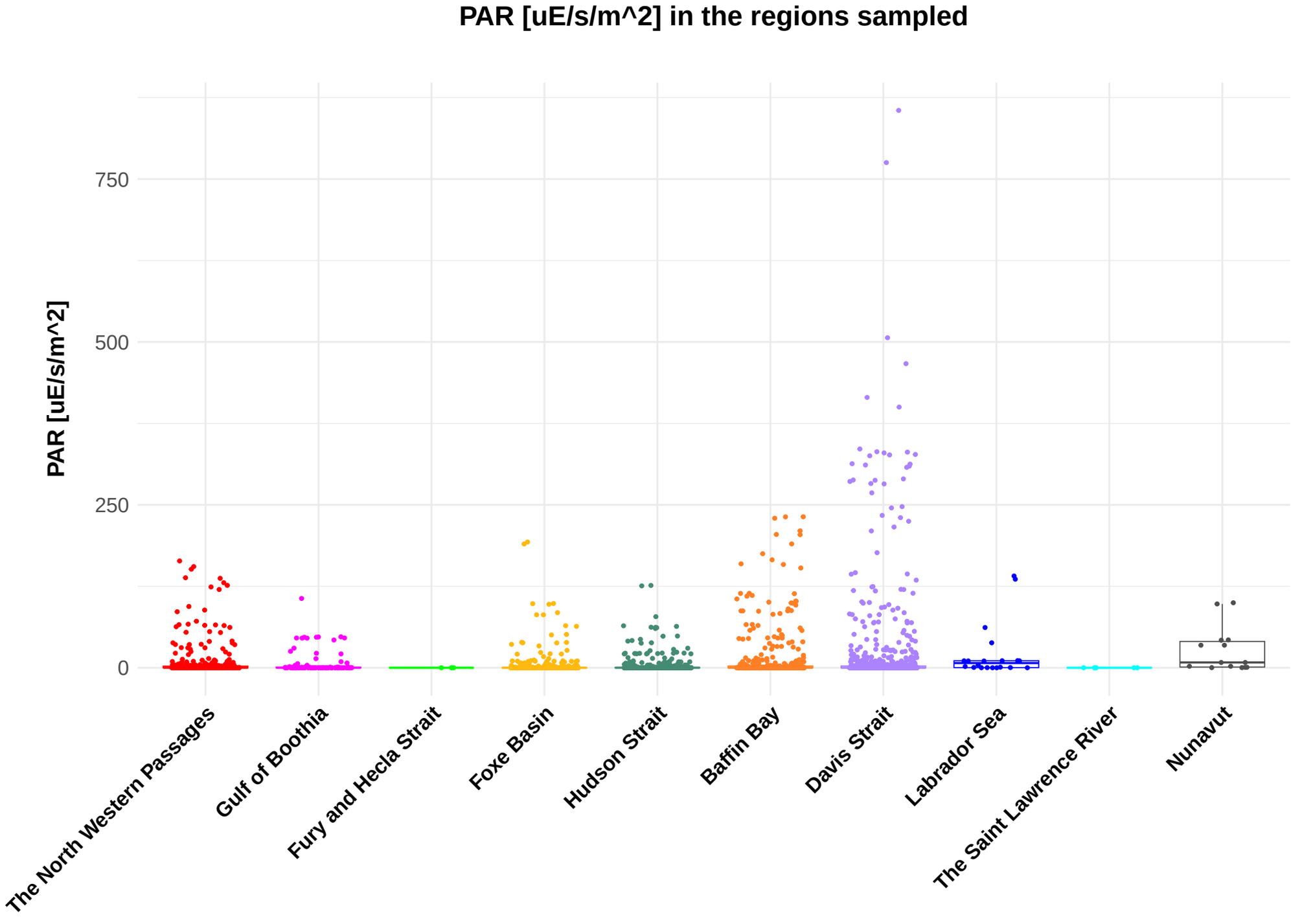

Sigma−T [kg/m^3] in the regions sampled
Sigma−T [kg/m^3]
28
26
24
22
20
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

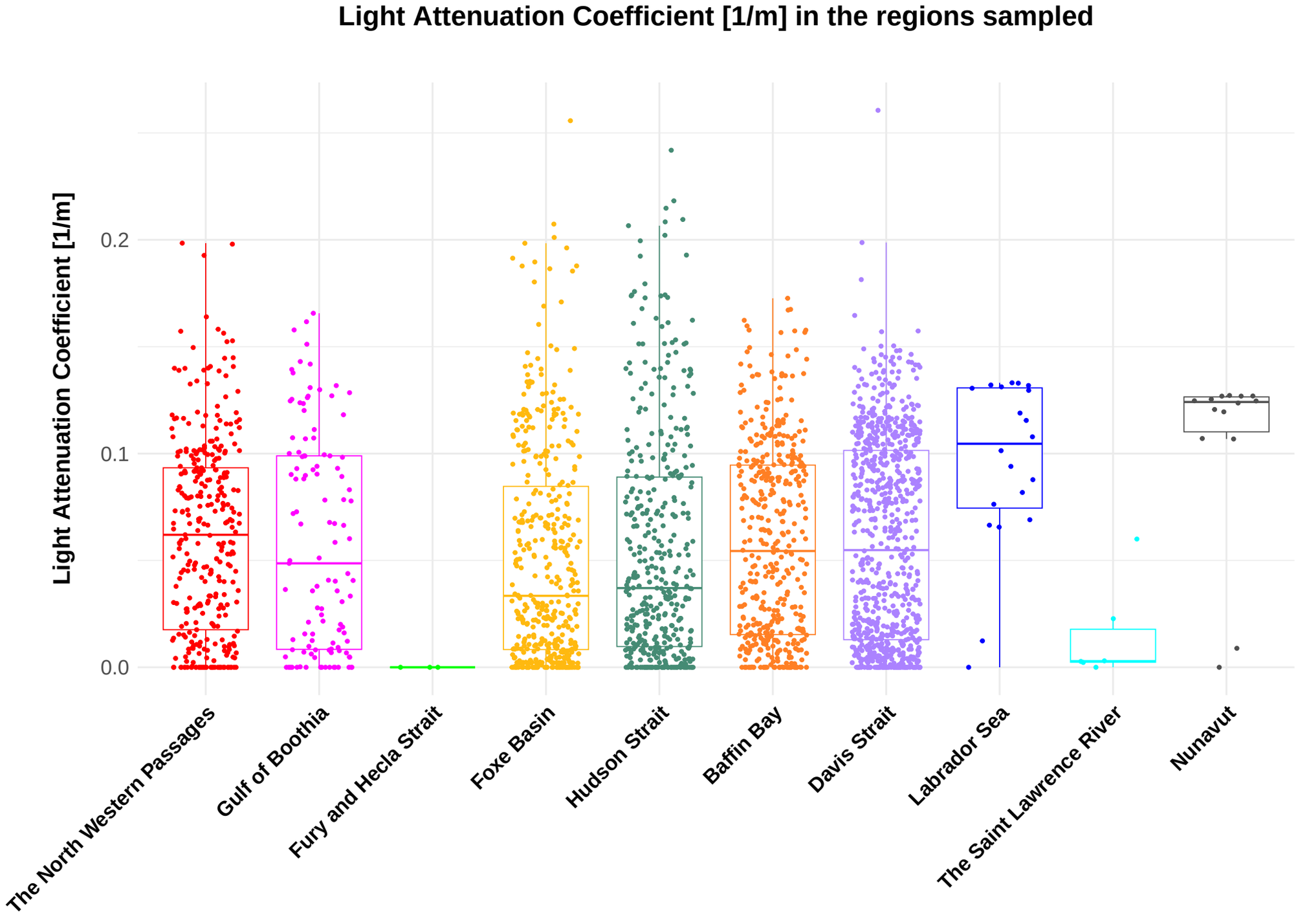

Sigma−Theta [kg/m^3] in the regions sampled
Sigma−Theta [kg/m^3]
28
26
24
22
20
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

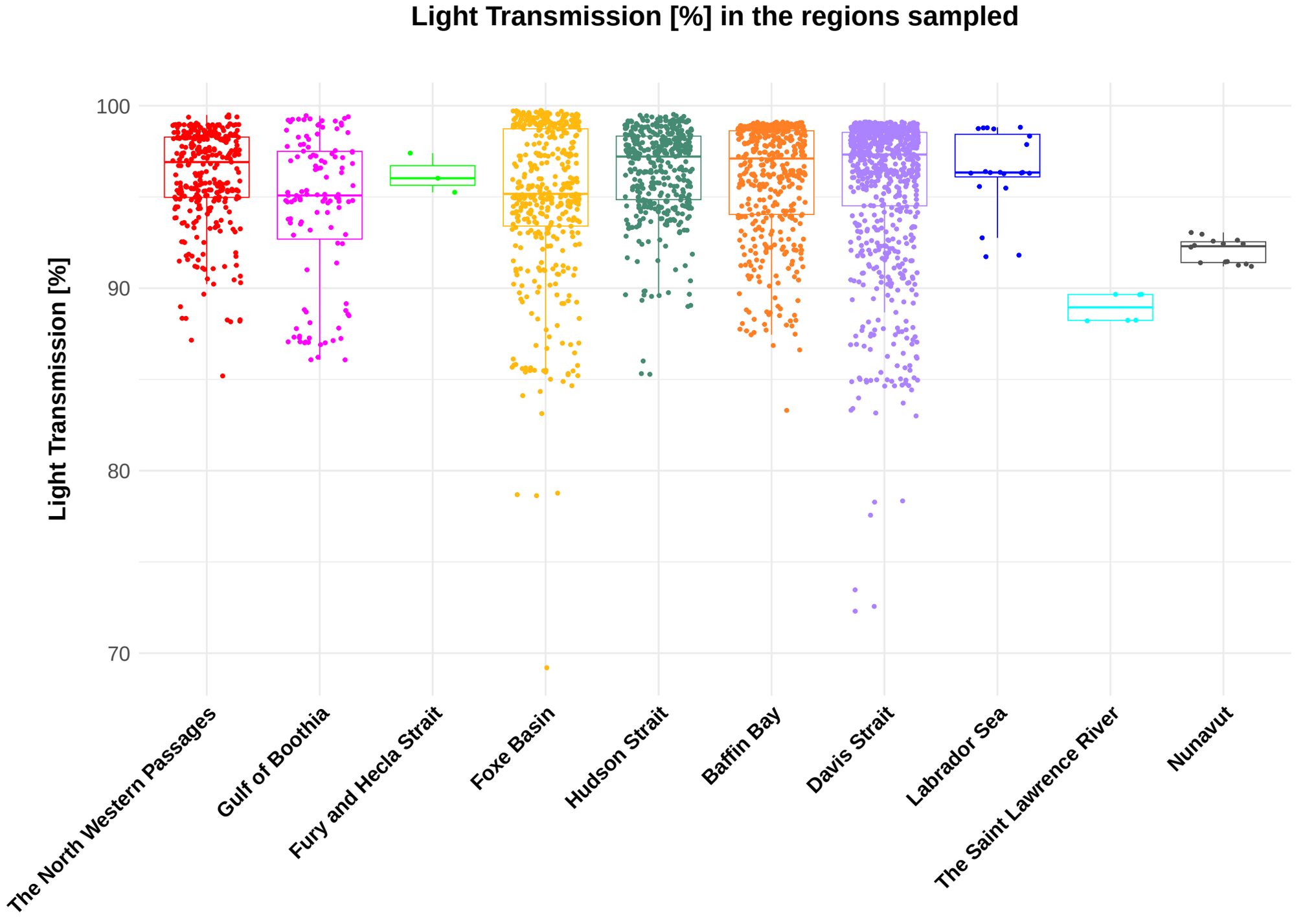

Sound Velocity [m/s] in the regions sampled
Sound Velocity [m/s]
1480
1470
1460
1450
1440
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

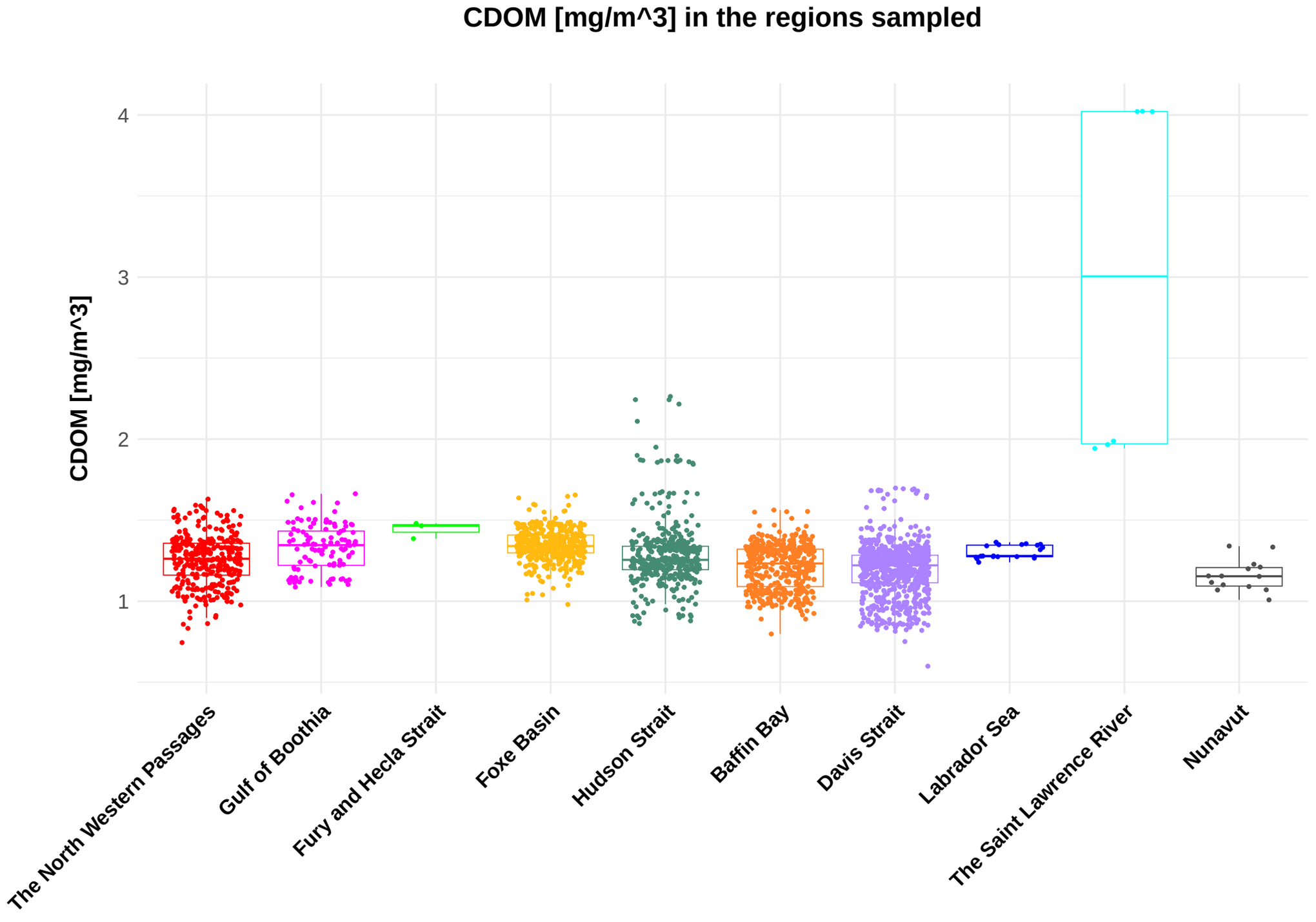

Dissolved Oxygen [uM] in the regions sampled
Dissolved Oxygen [uM]
400
300
200
100
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

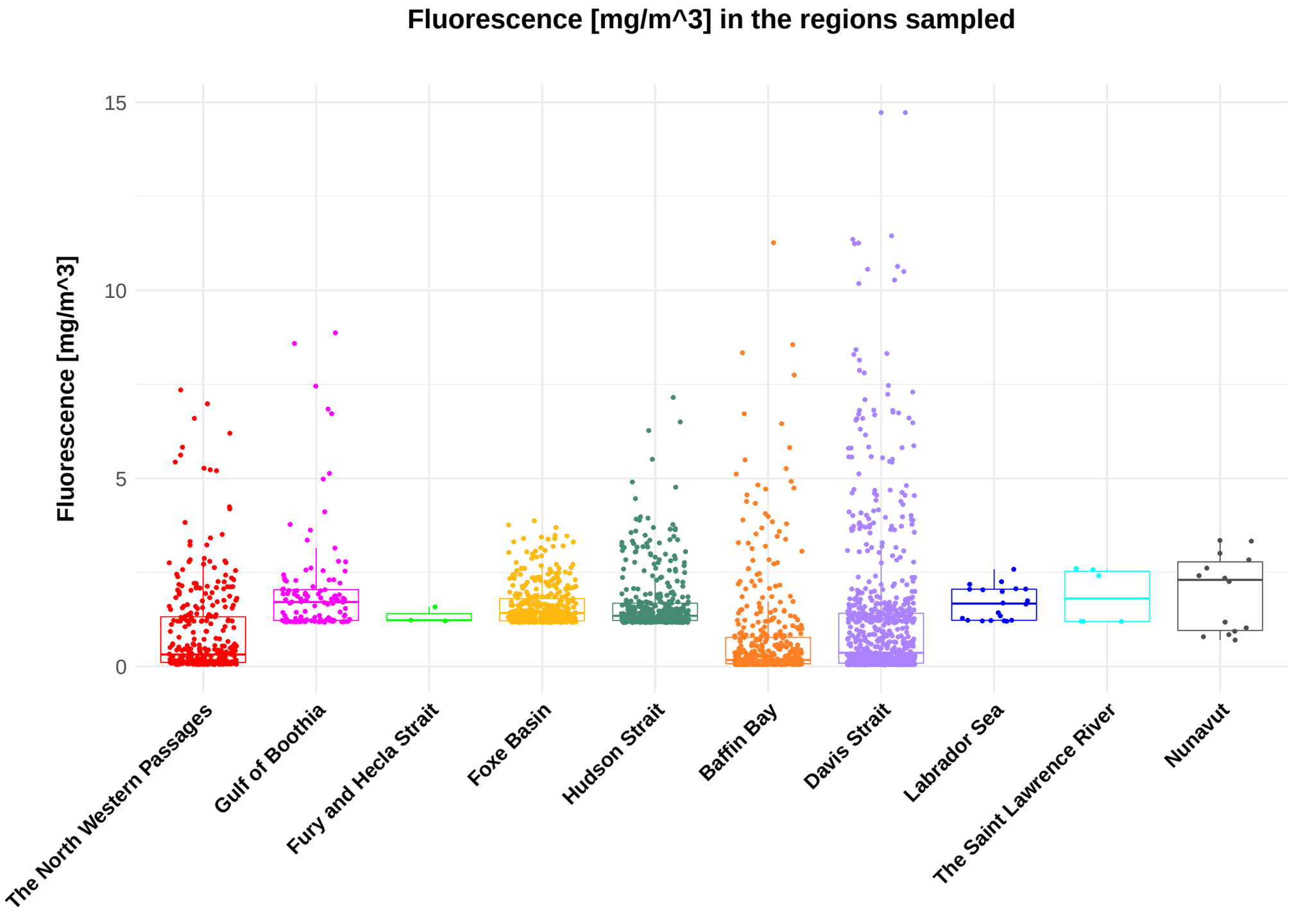

Oxygen solubility [ml/l] in the regions sampled
Oxygen solubility [ml/l]
8.5
8.0
7.5
7.0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

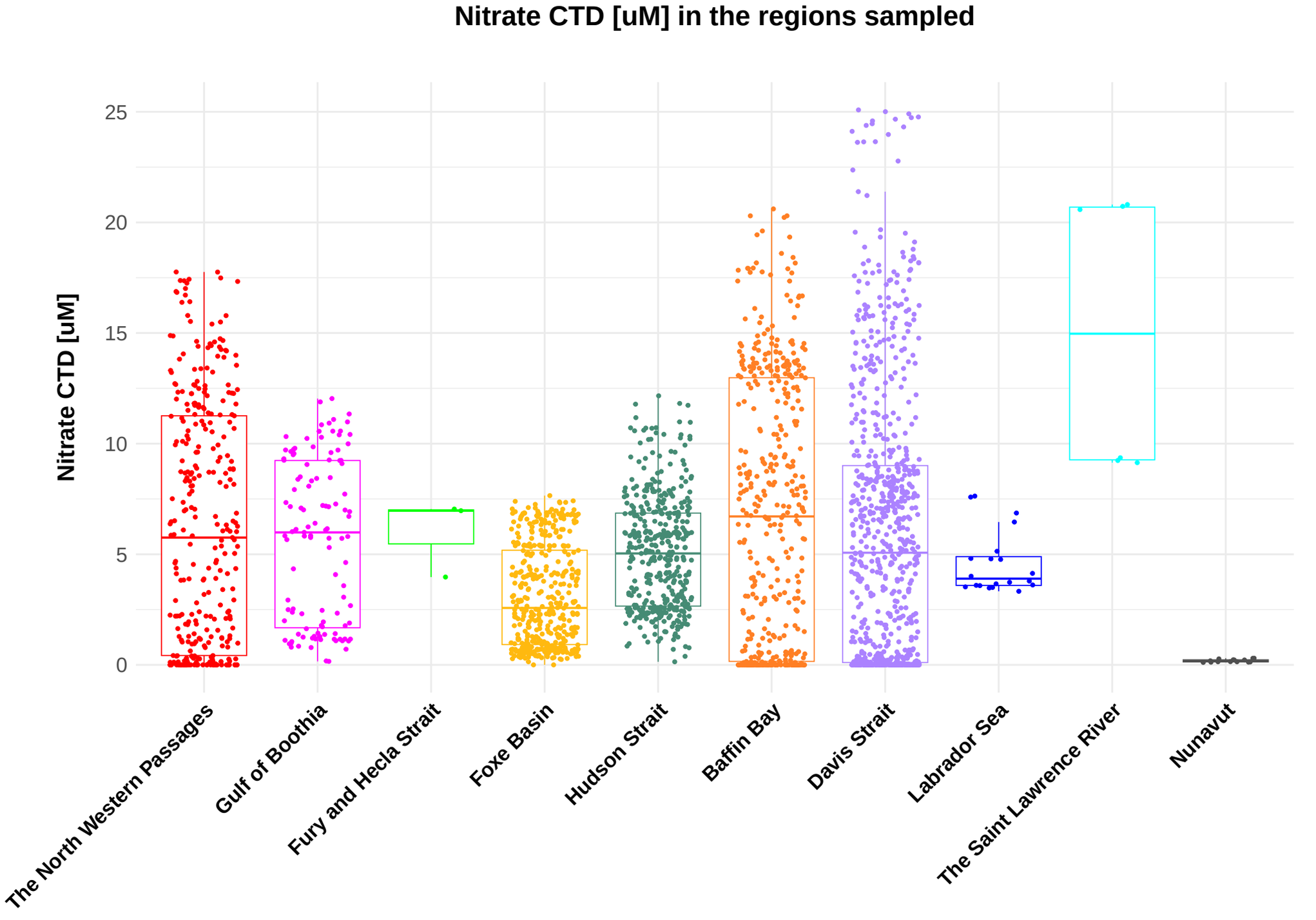

PAR [uE/s/m^2] in the regions sampled
PAR [uE/s/m^2]
750
500
250
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

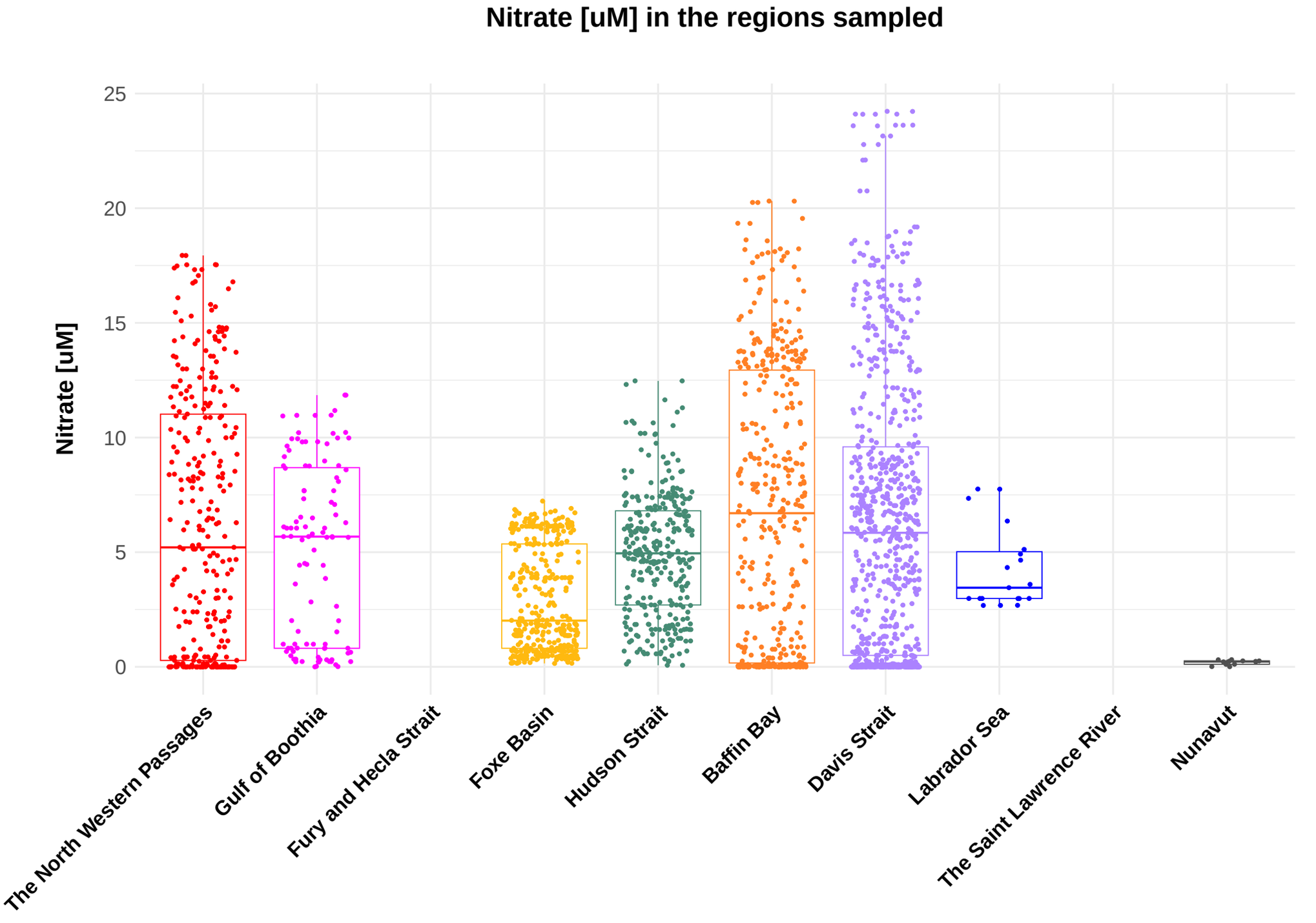

Light Attenuation Coefficient [1/m] in the regions sampled
Light Attenuation Coefficient [1/m]
0.2
0.1
0.0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

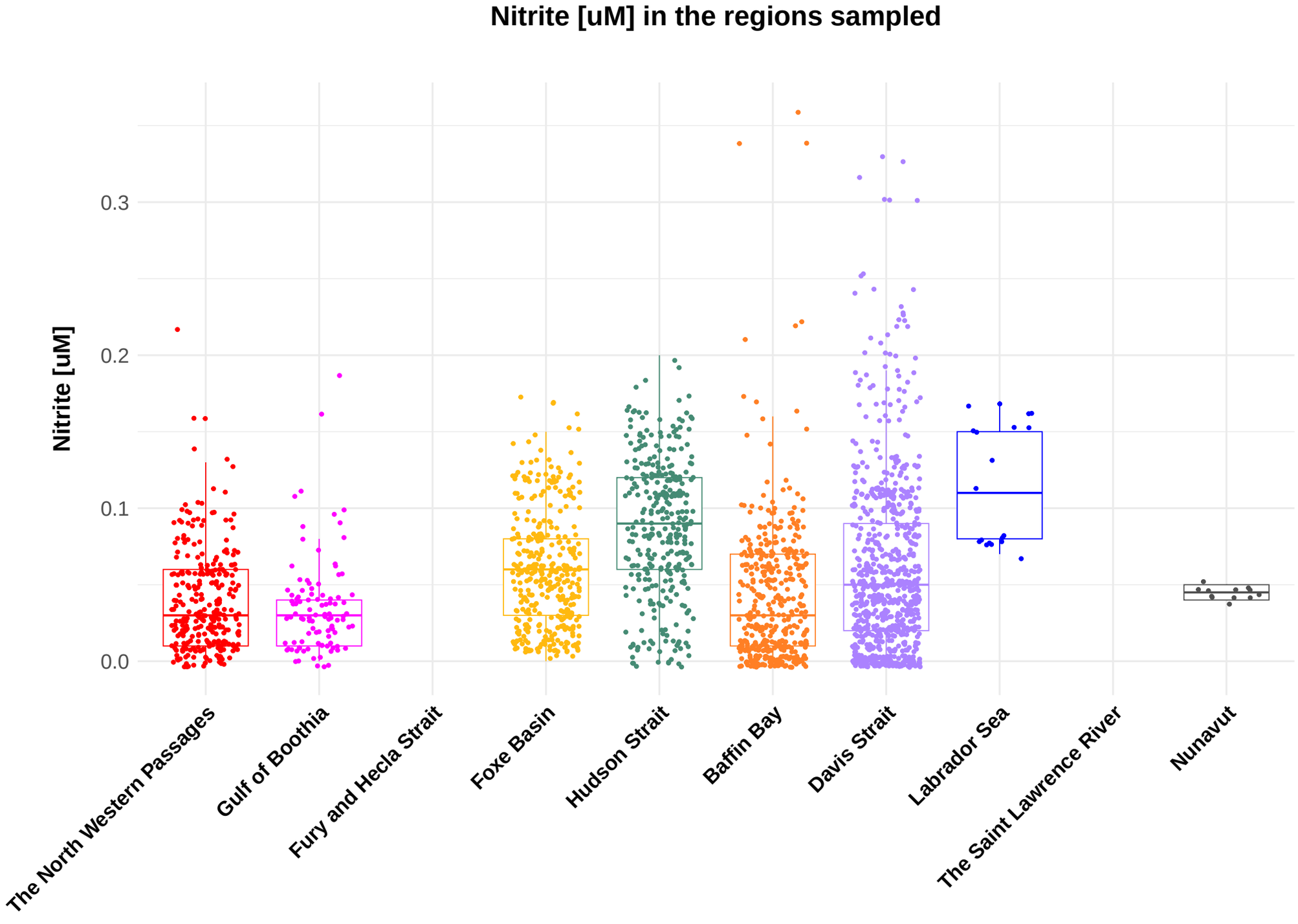

Light Transmission [%] in the regions sampled
Light Transmission [%]
100
90
80
70
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

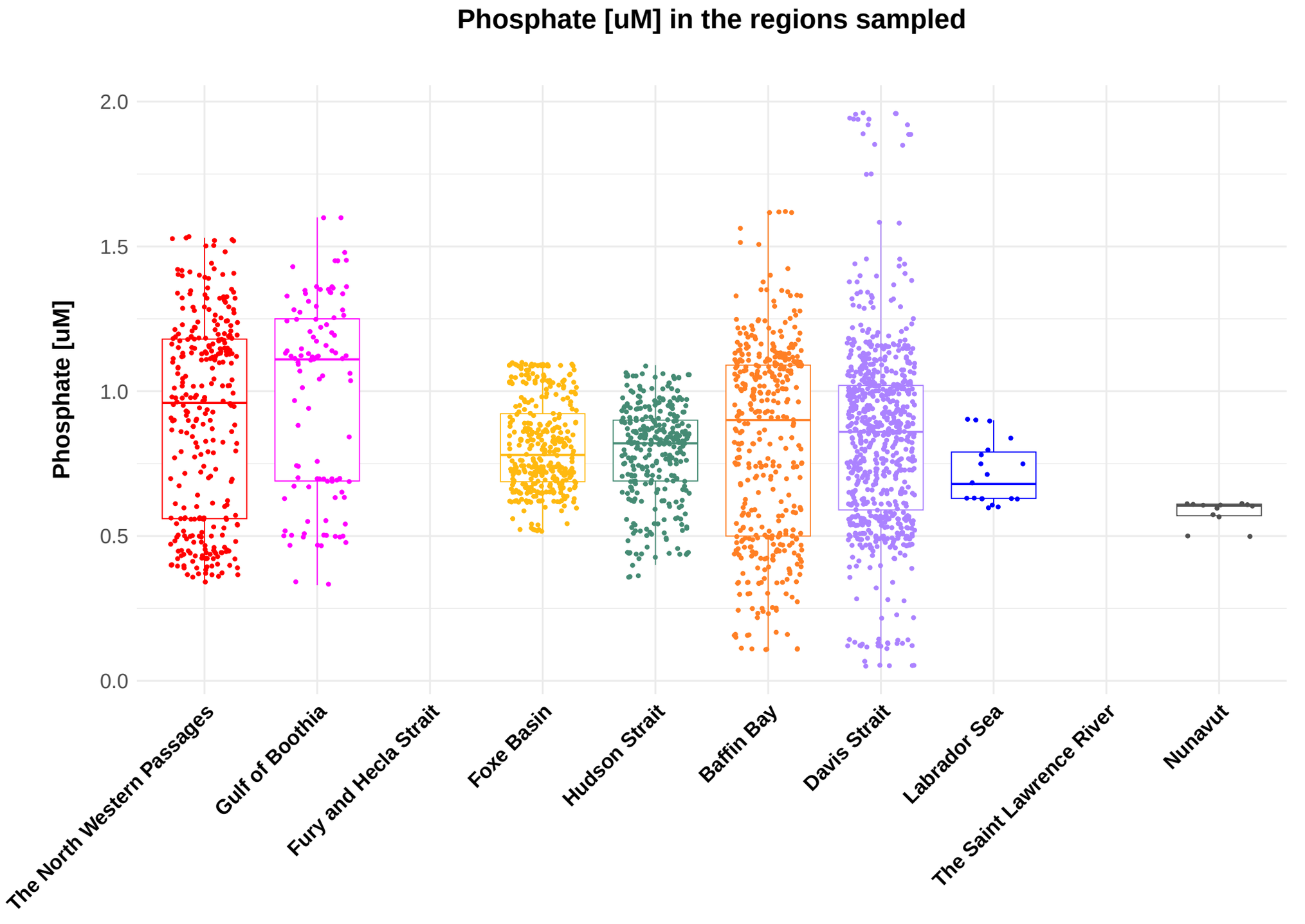

CDOM [mg/m^3] in the regions sampled
CDOM [mg/m^3]
4
3
2
1
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

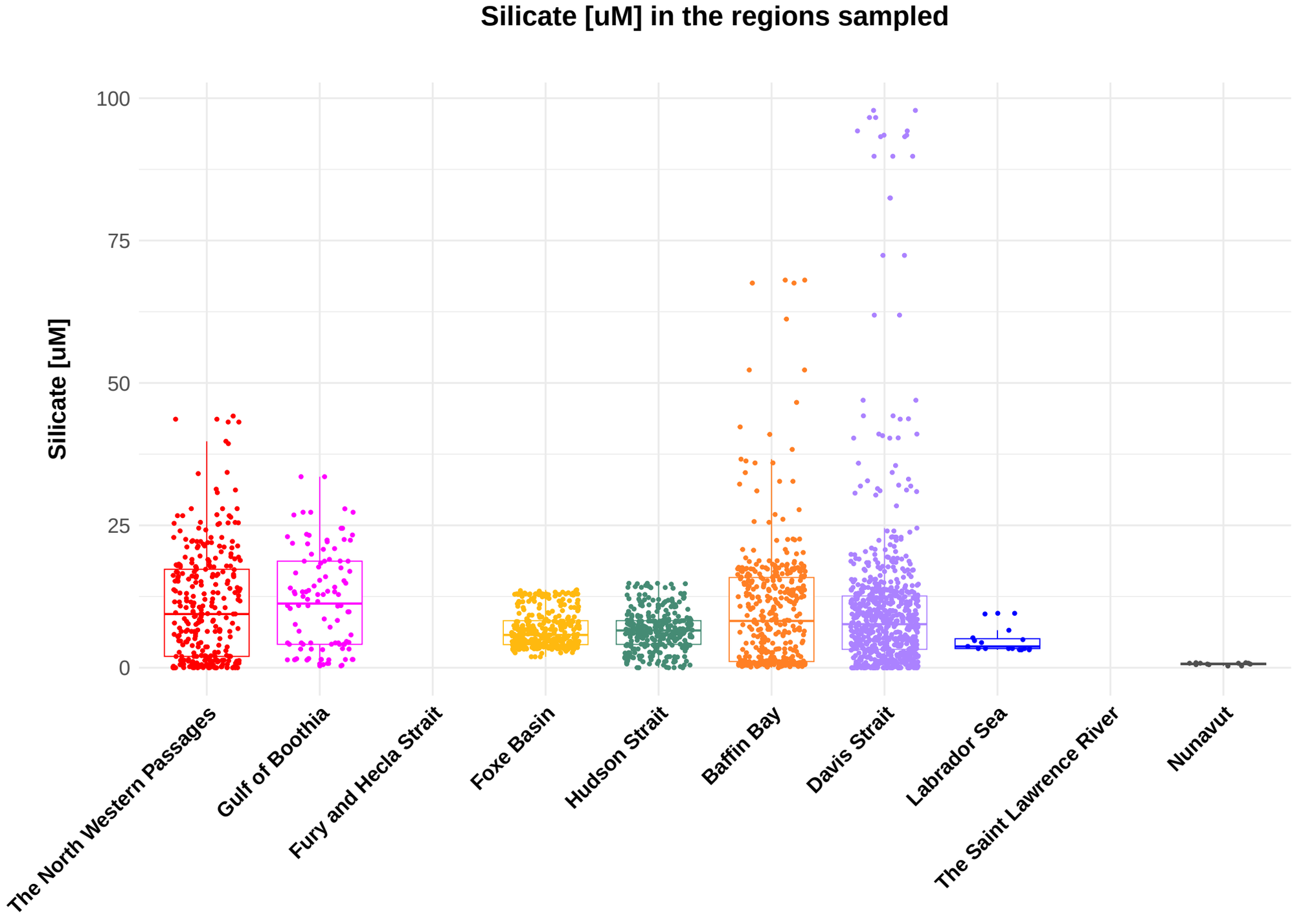

Fluorescence [mg/m^3] in the regions sampled
Fluorescence [mg/m^3]
15
10
5
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

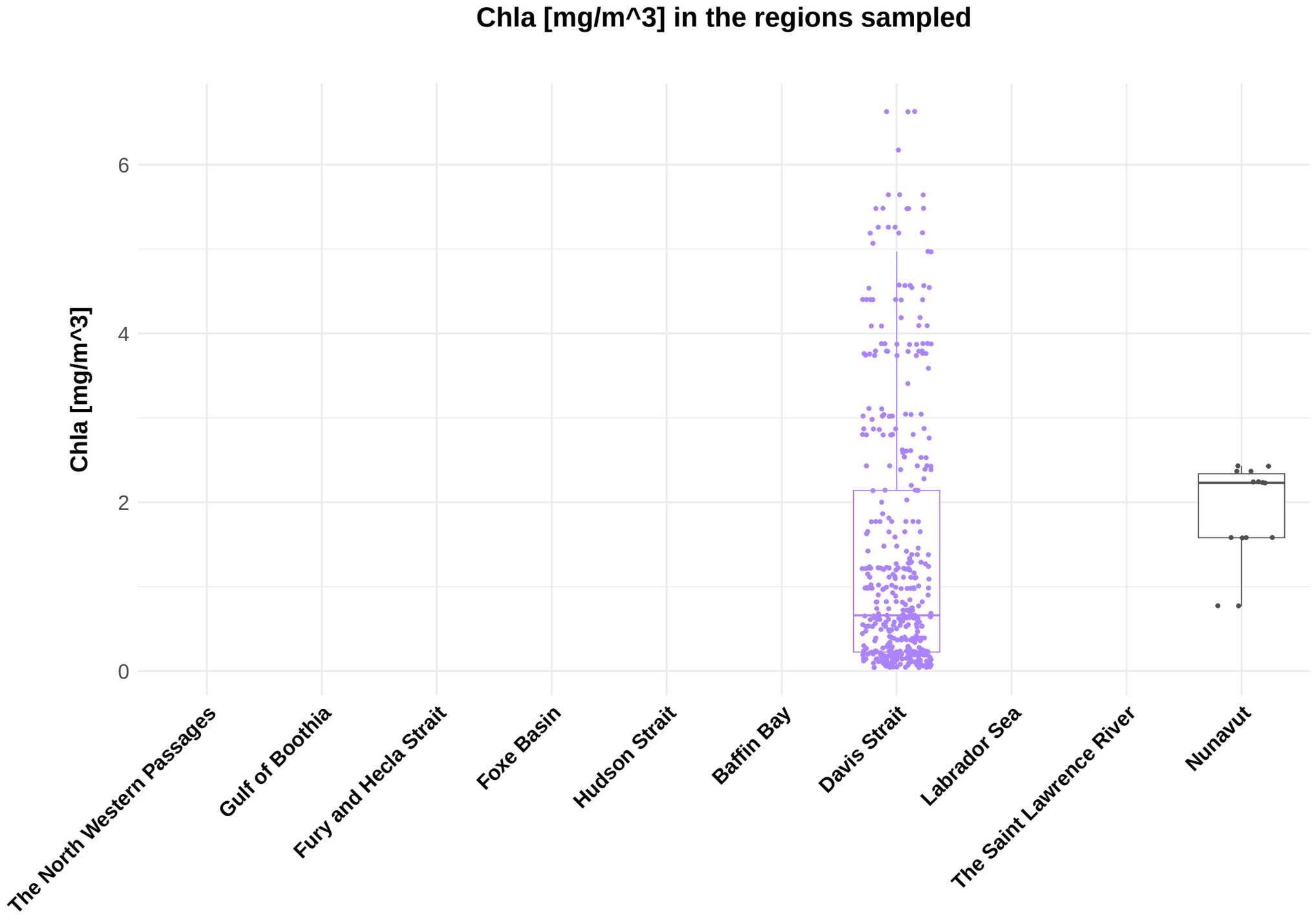

Nitrate CTD [uM] in the regions sampled
Nitrate CTD [uM]
25
20
15
10
5
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

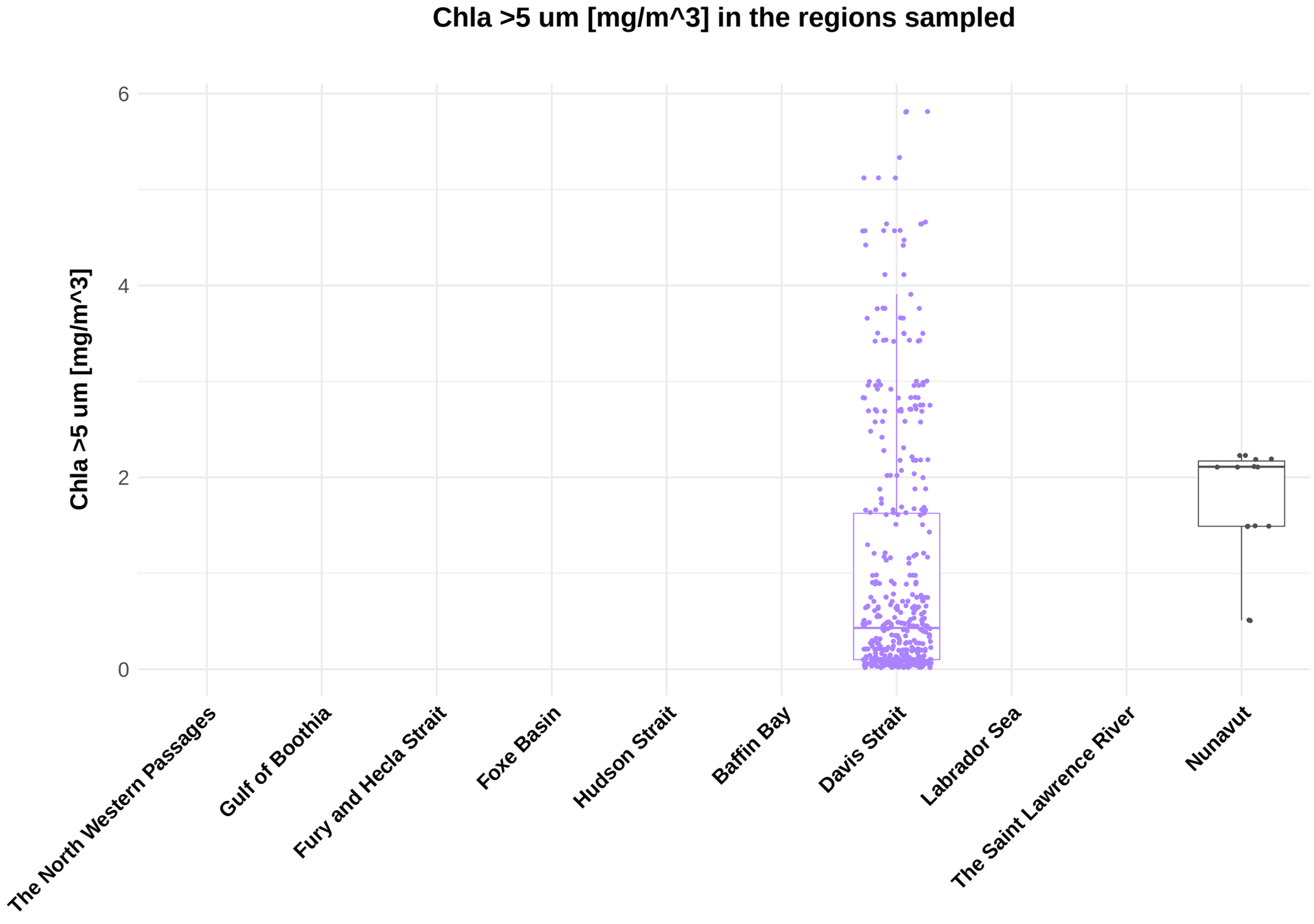

Nitrate [uM] in the regions sampled
Nitrate [uM]
25
20
15
10
5
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

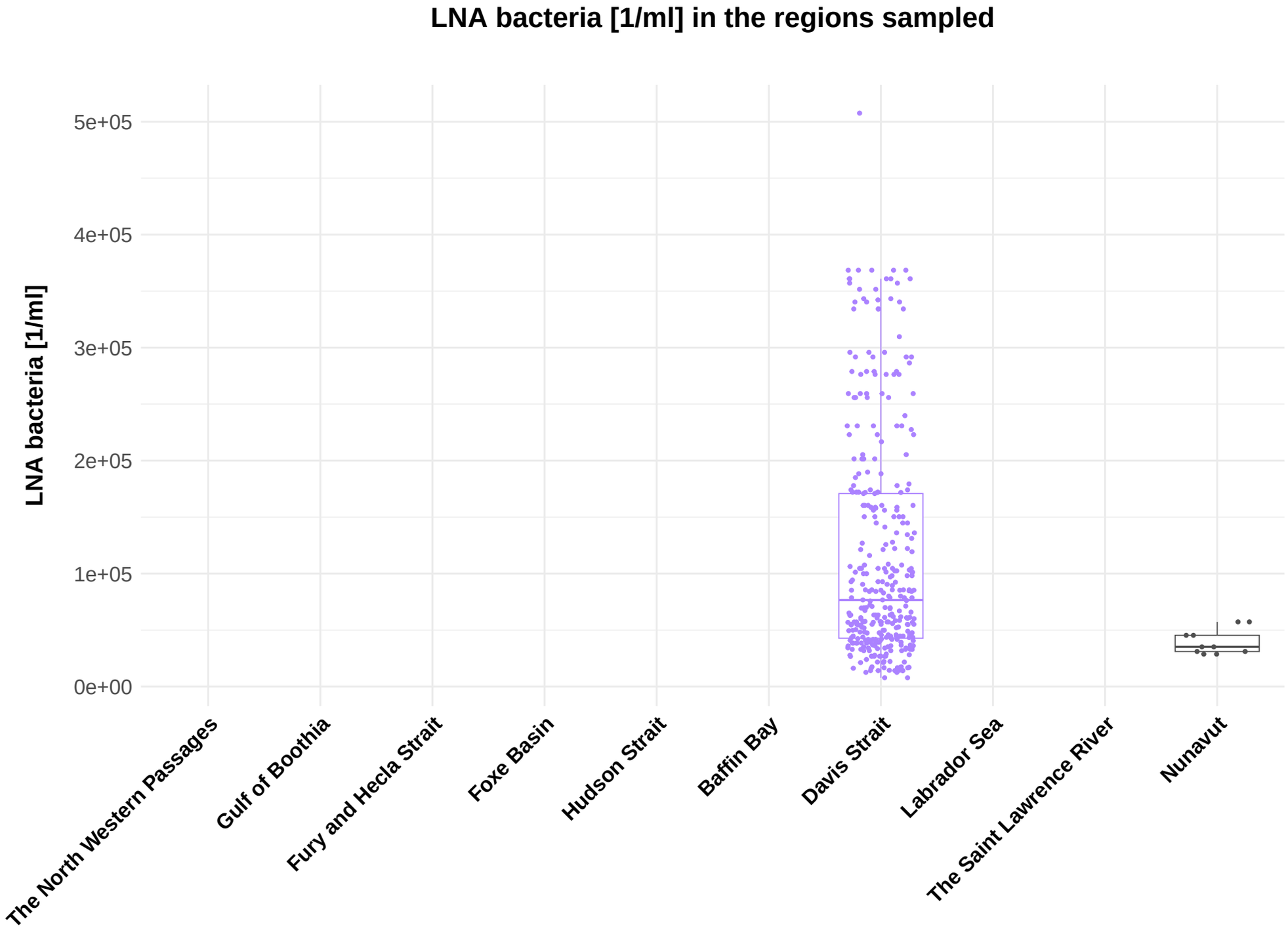

Nitrite [uM] in the regions sampled
Nitrite [uM]
0.0
0.1
0.2
0.3
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

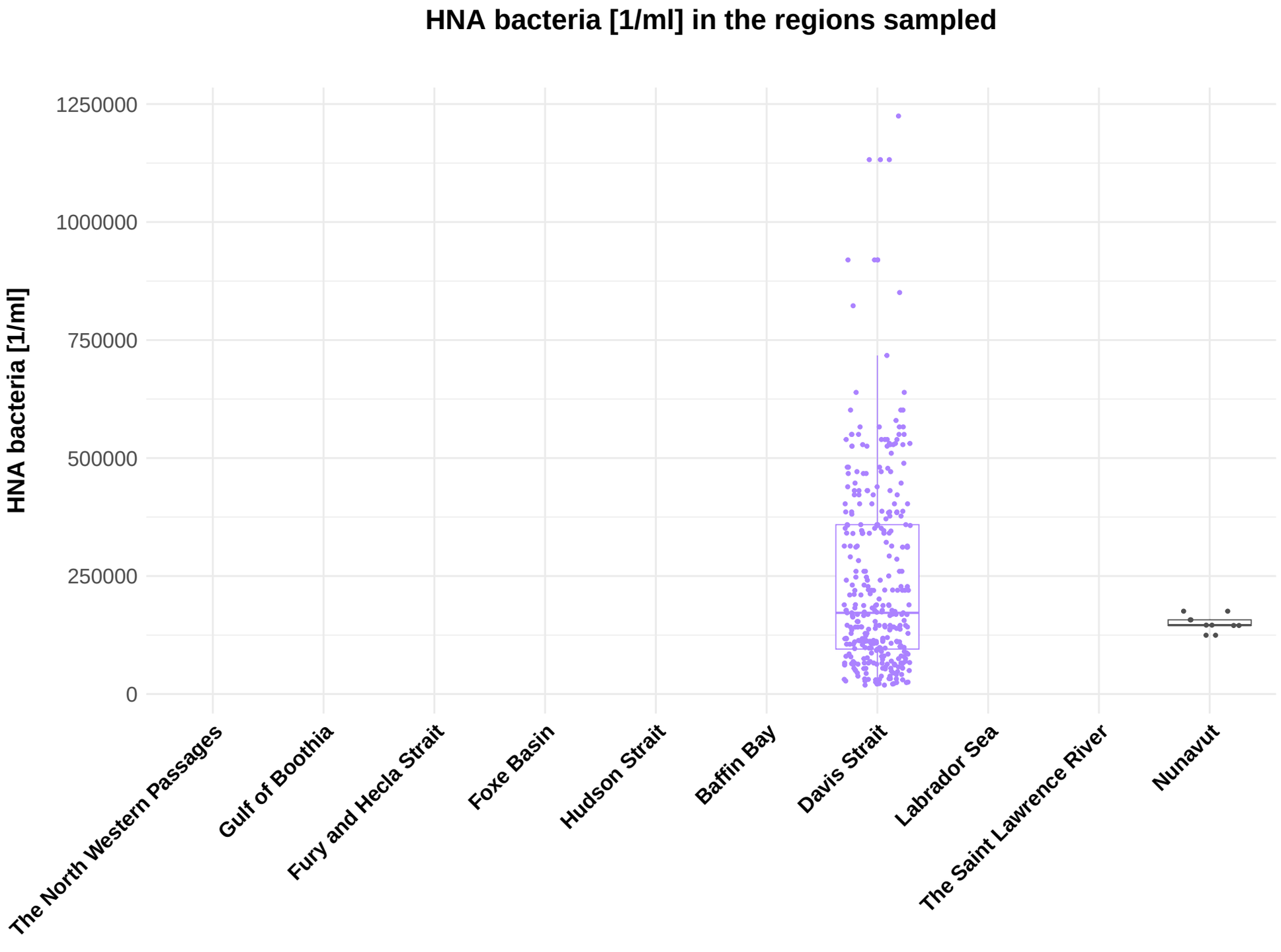

Phosphate [uM] in the regions sampled
Phosphate [uM]
2.0
1.5
1.0
0.5
0.0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

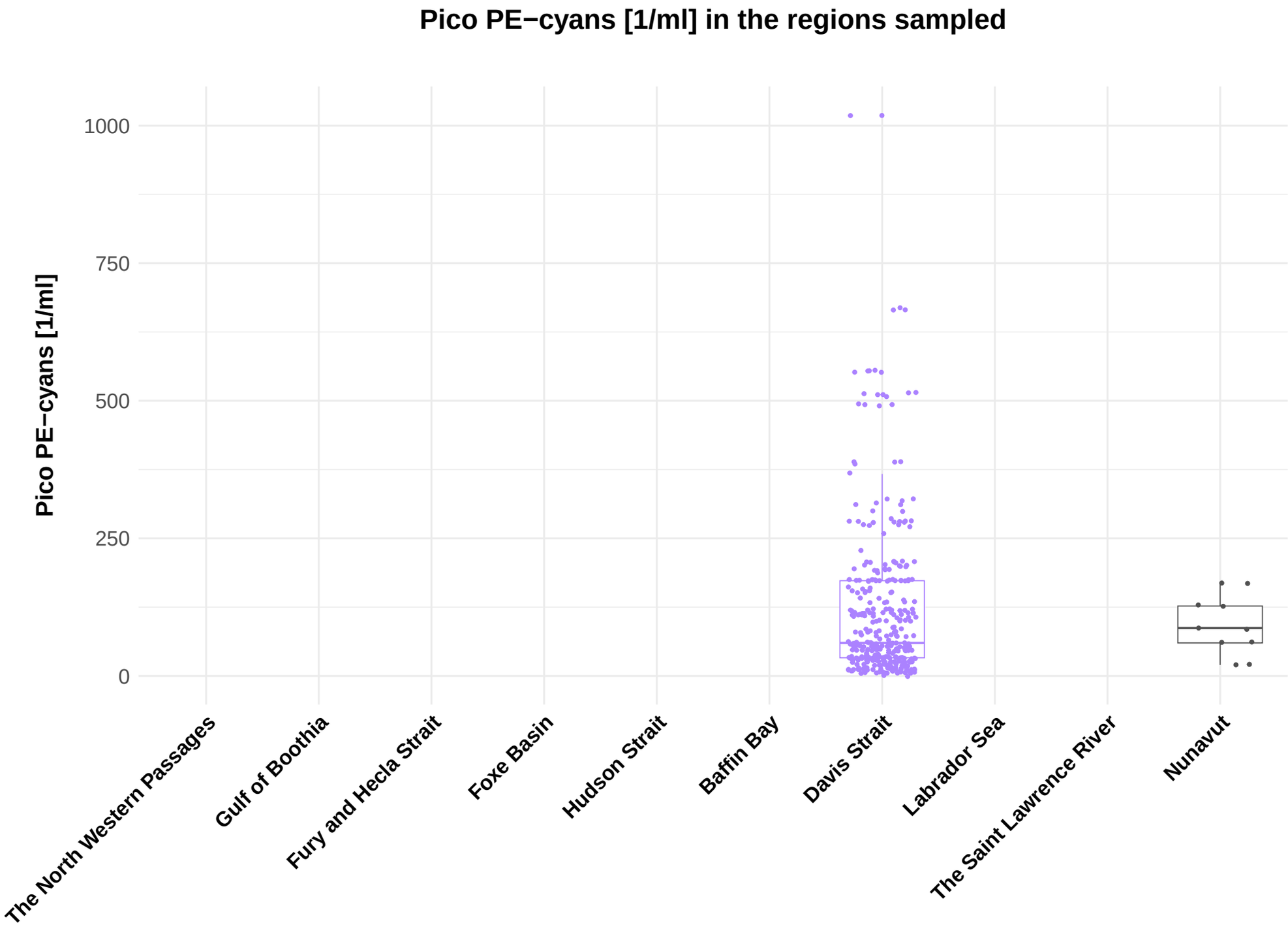

Silicate [uM] in the regions sampled
Silicate [uM]
100
75
50
25
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

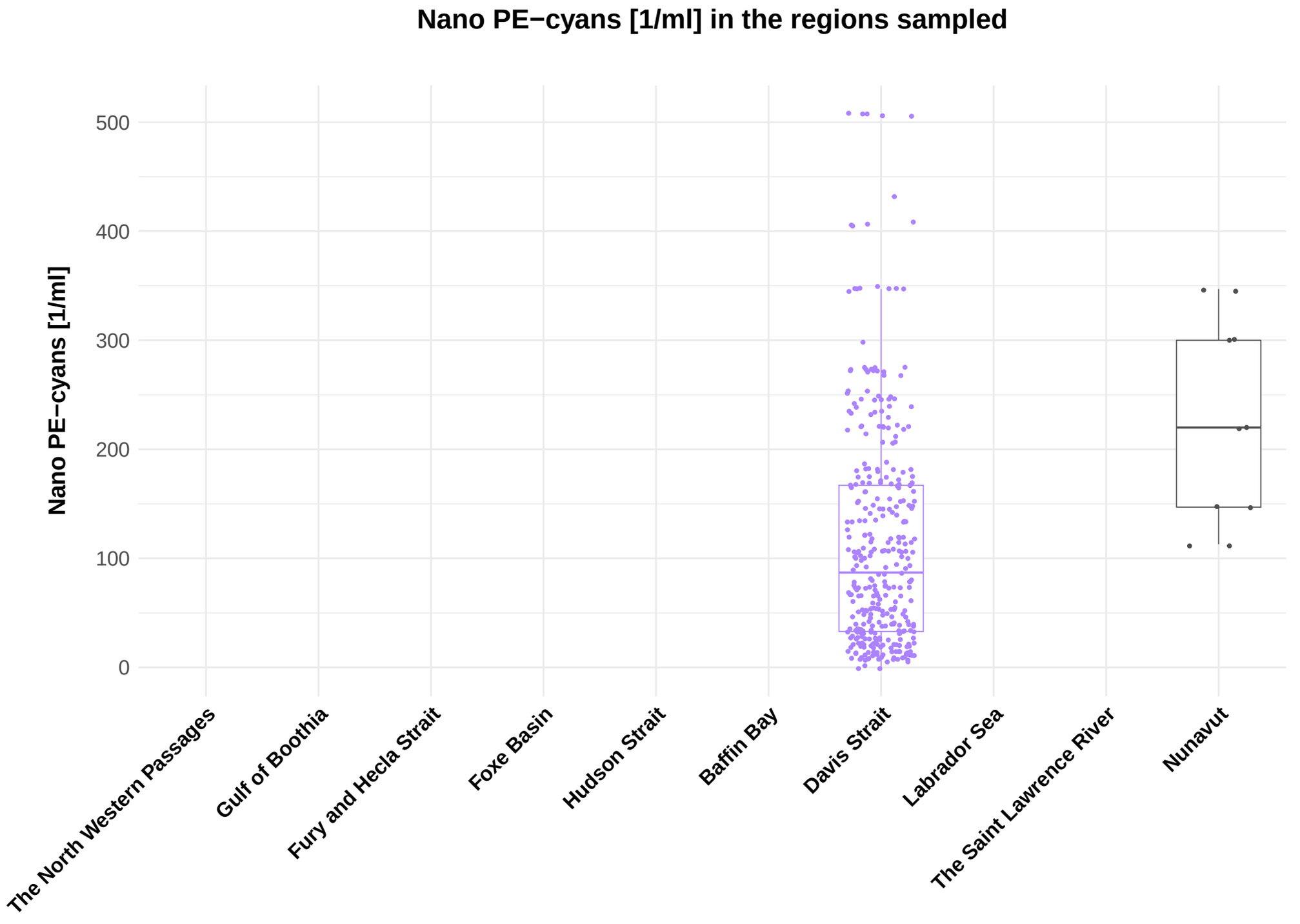

Chla [mg/m^3] in the regions sampled
Chla [mg/m^3]
0
2
4
6
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

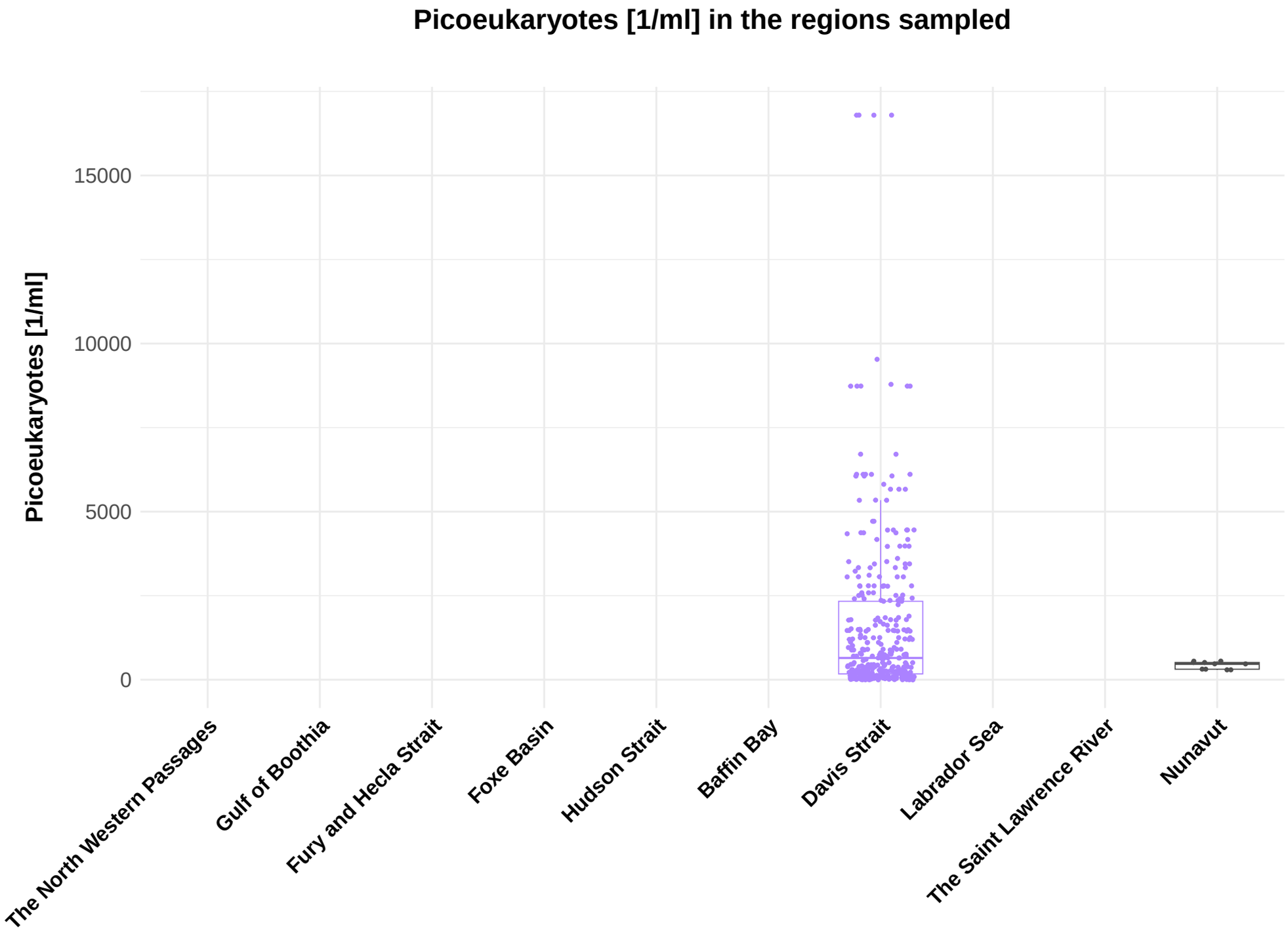

Chla >5 um [mg/m^3] in the regions sampled
Chla >5 um [mg/m^3]
6
4
2
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

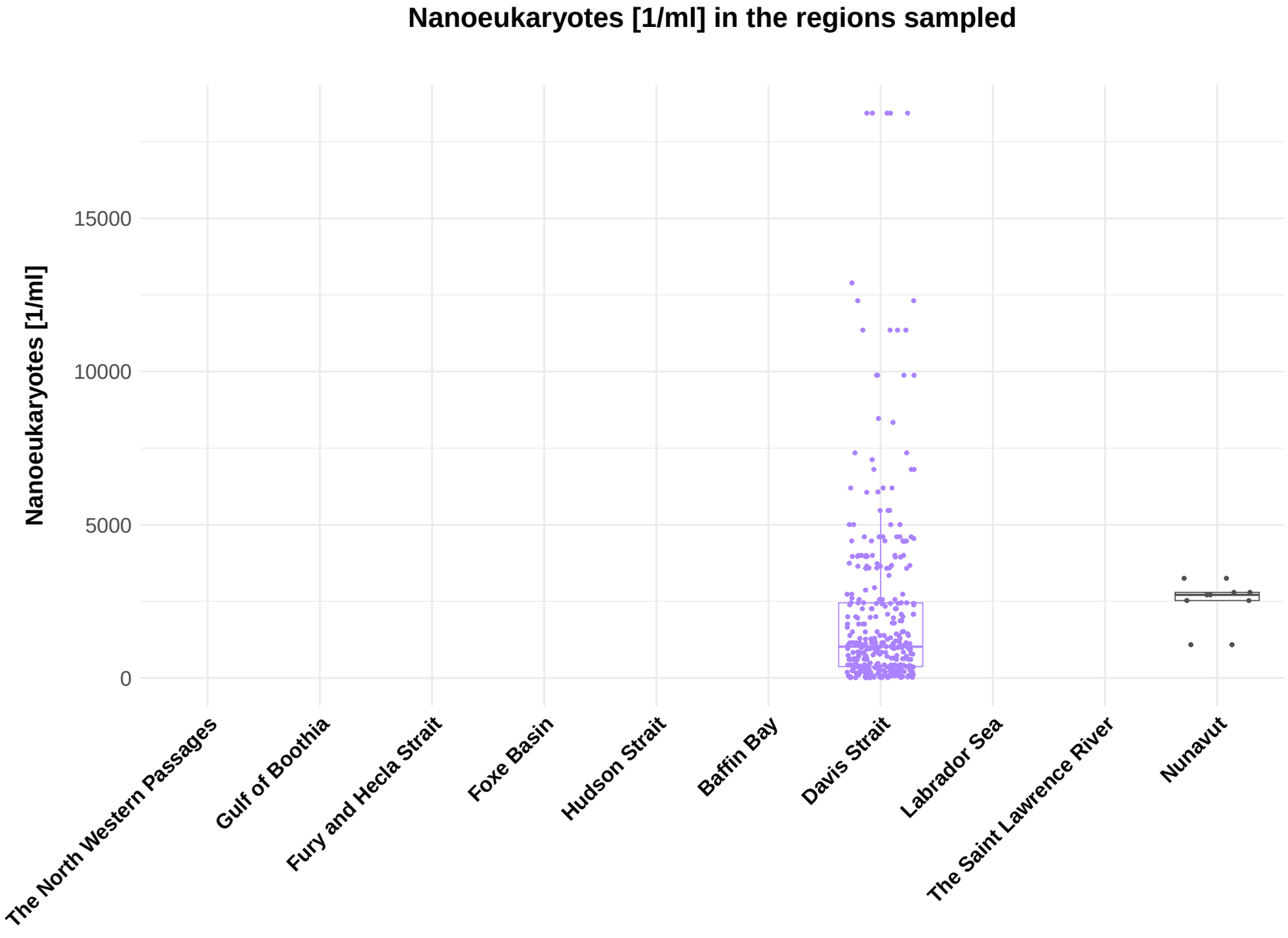

LNA bacteria [1/ml] in the regions sampled
LNA bacteria [1/ml]
5e+05
4e+05
3e+05
2e+05
1e+05
0e+00
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

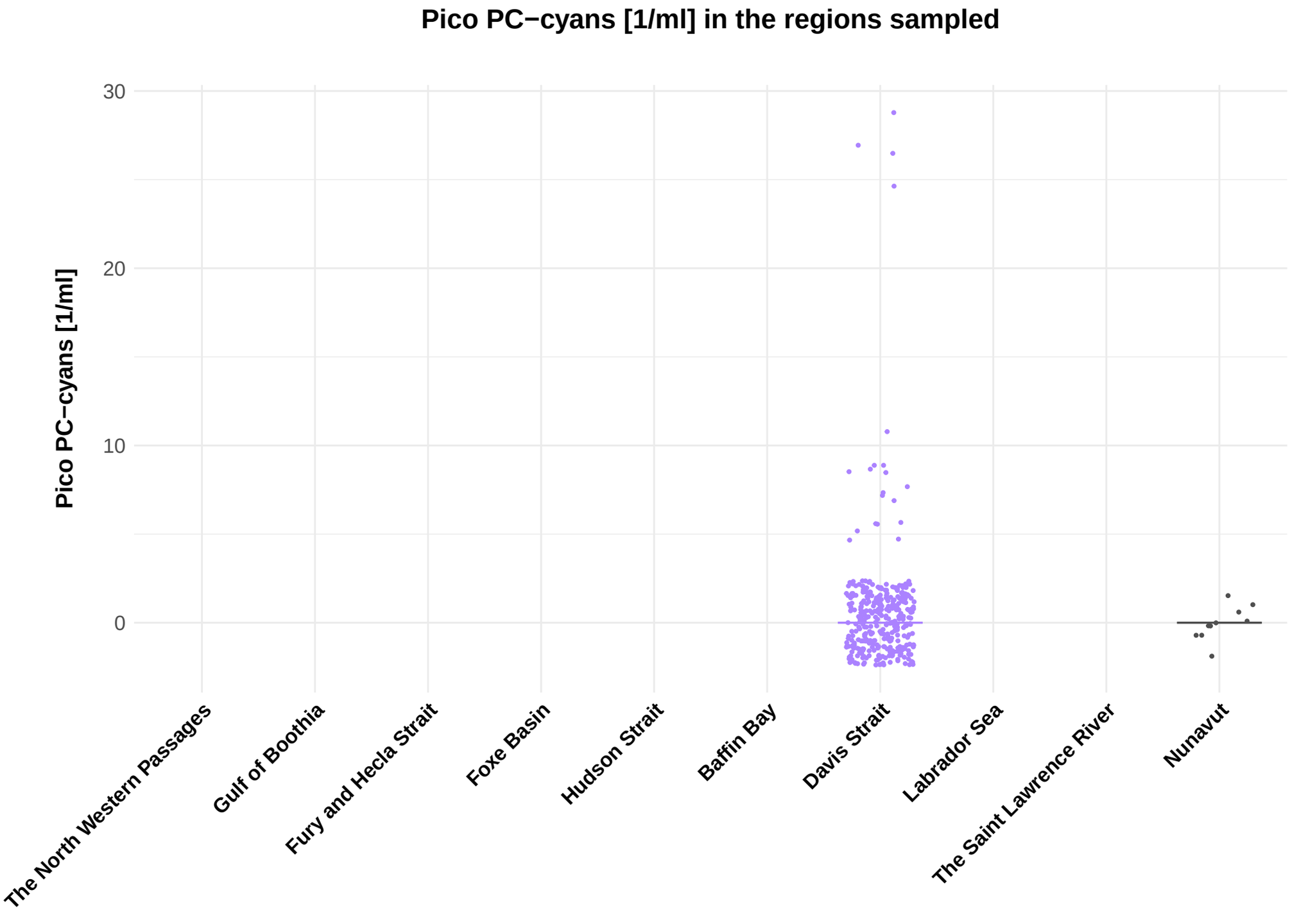

HNA bacteria [1/ml] in the regions sampled
HNA bacteria [1/ml]
1250000
1000000
750000
500000
250000
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

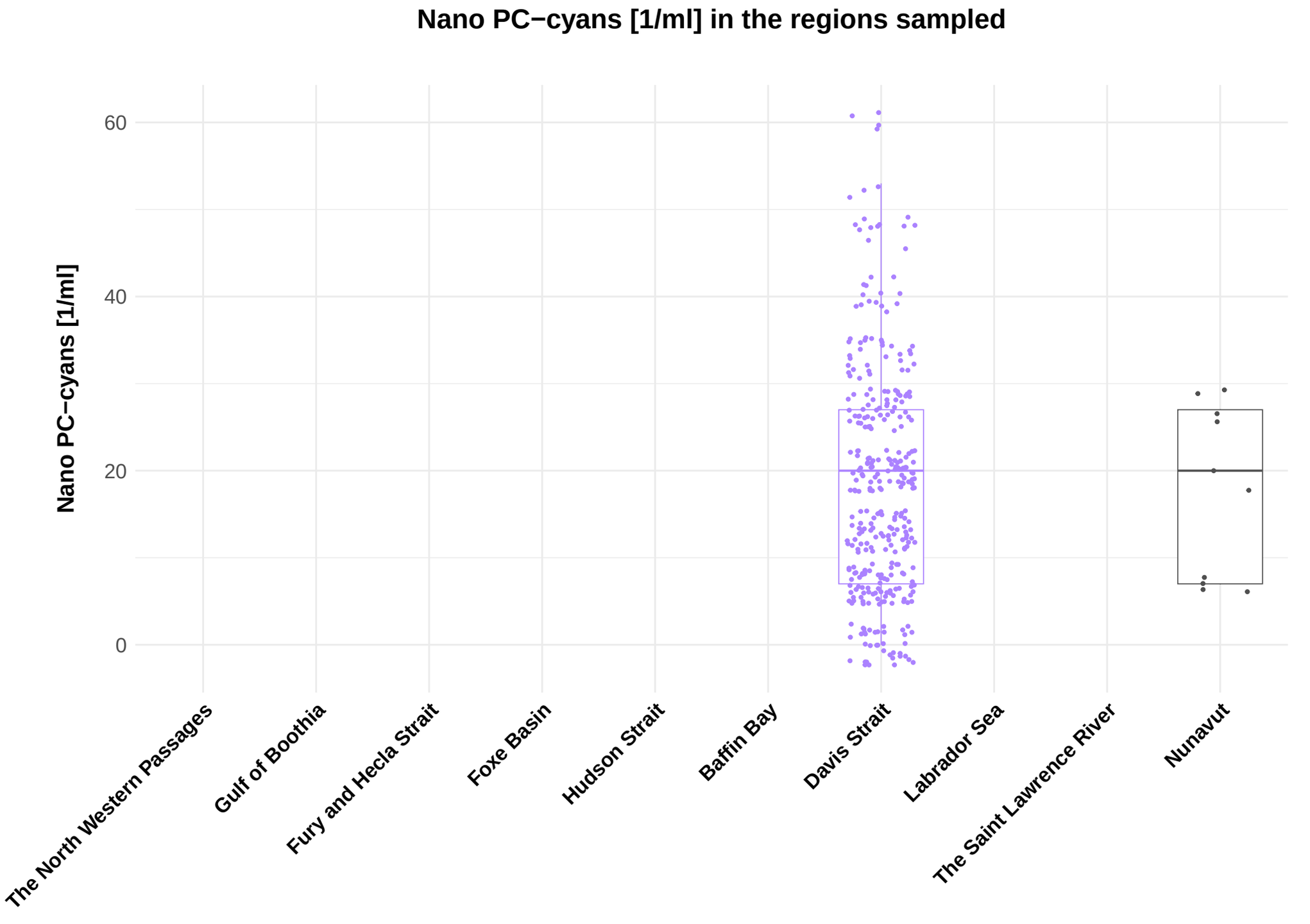

Pico PE−cyans [1/ml] in the regions sampled
Pico PE−cyans [1/ml]
1000
750
500
250
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

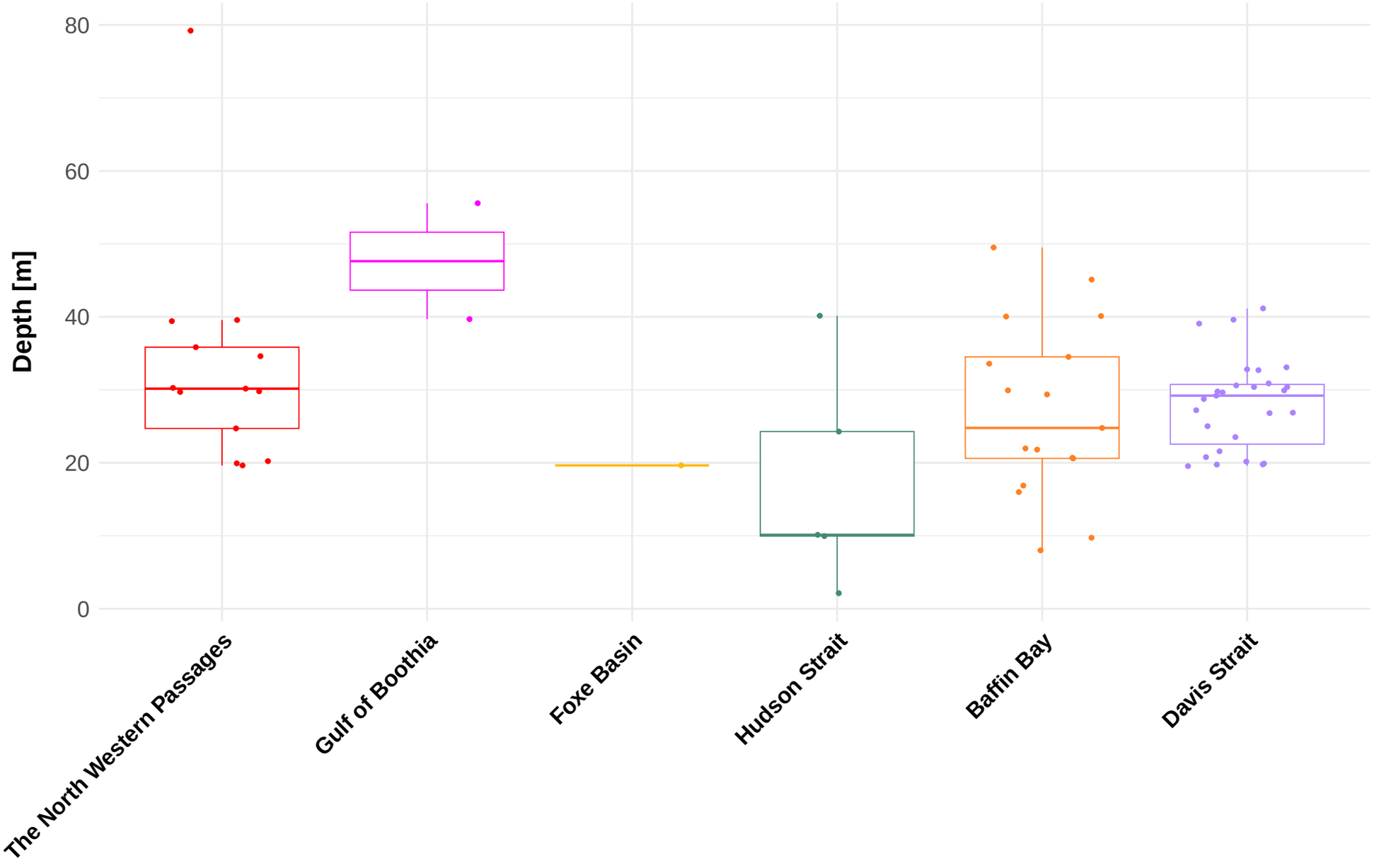

Nano PE−cyans [1/ml] in the regions sampled
Nano PE−cyans [1/ml]
500
400
300
200
100
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

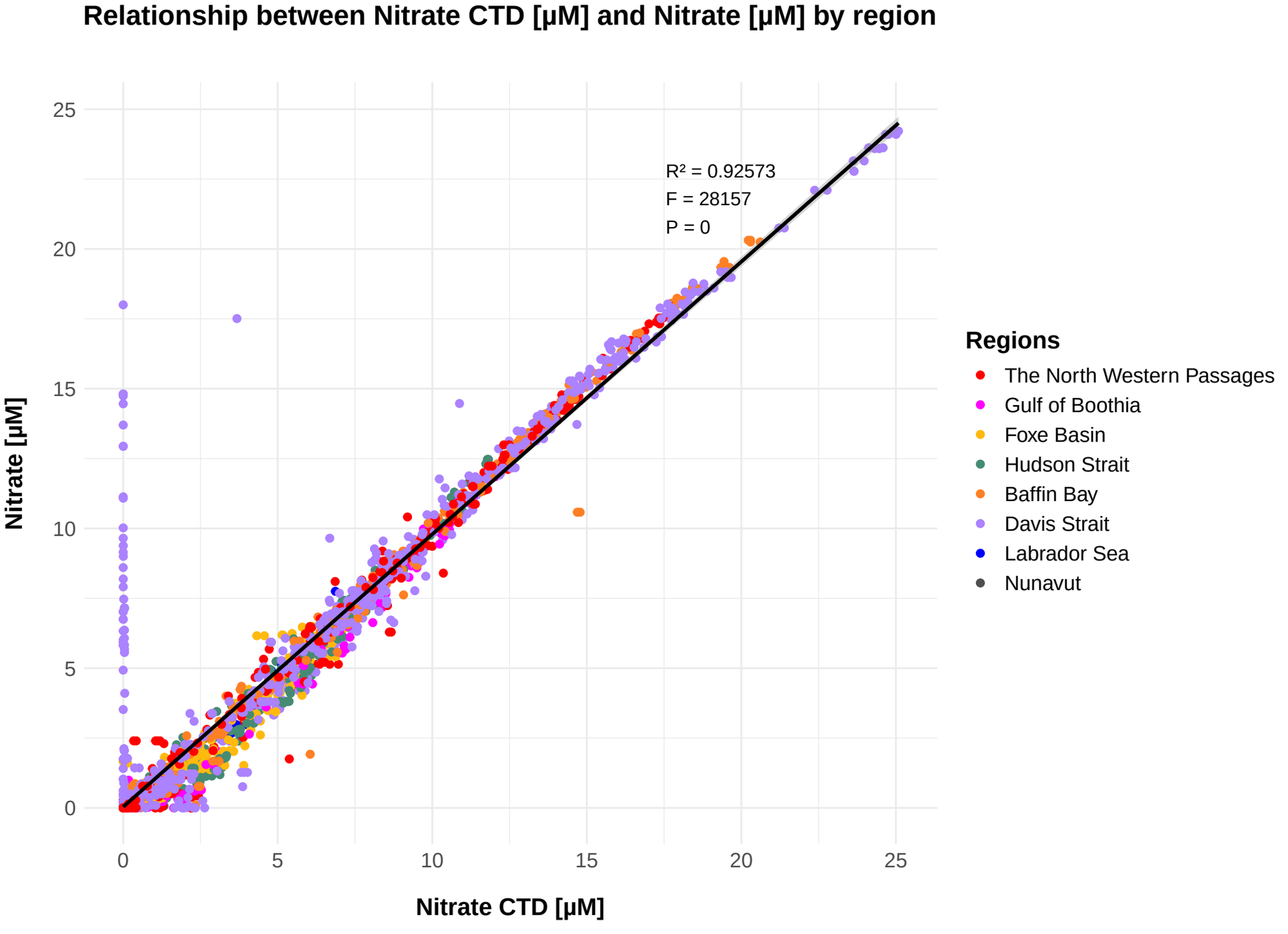

Picoeukaryotes [1/ml] in the regions sampled
Picoeukaryotes [1/ml]
15000
10000
5000
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

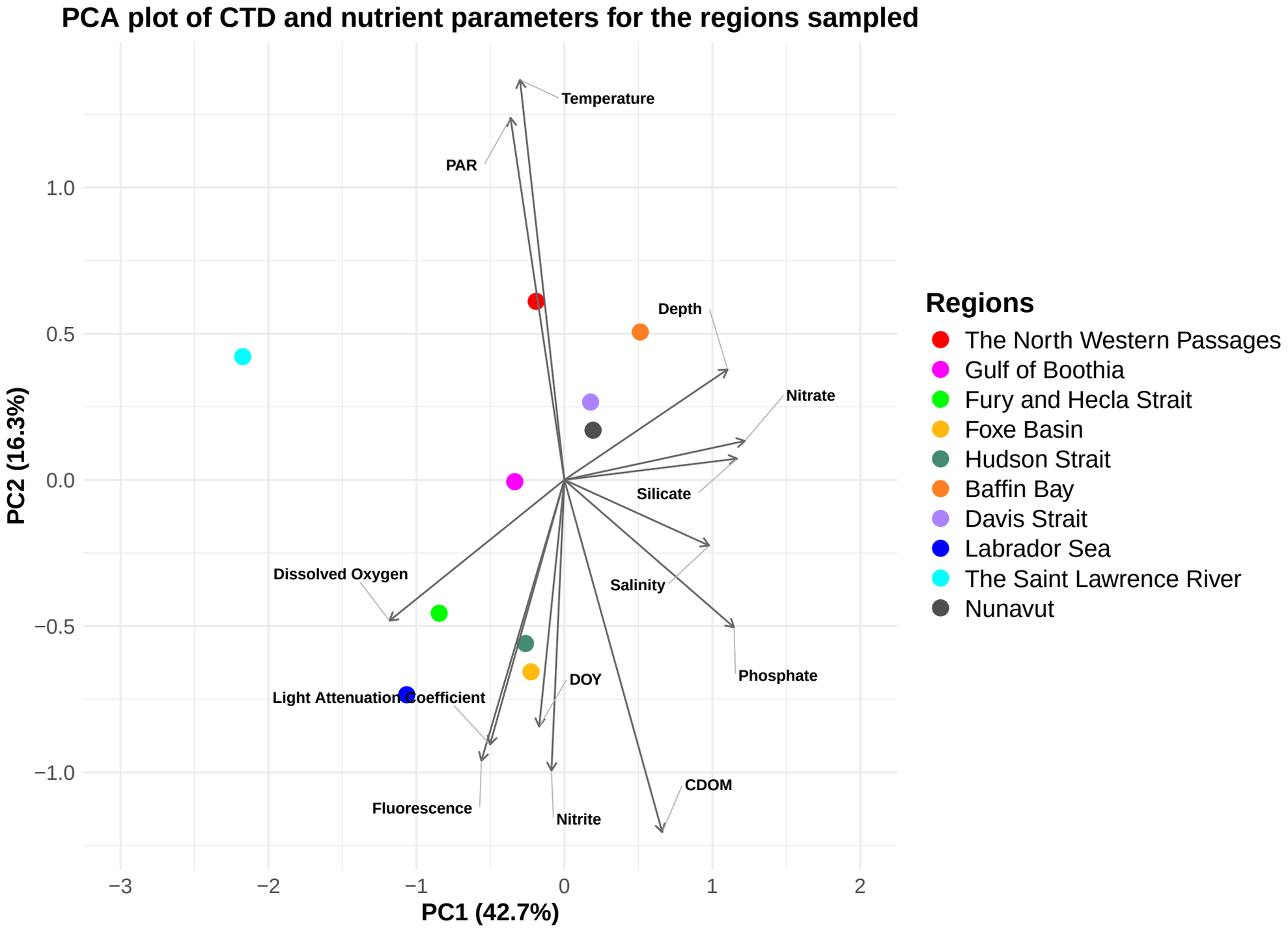

Nanoeukaryotes [1/ml] in the regions sampled
Nanoeukaryotes [1/ml]
15000
10000
5000
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut


Pico PC−cyans [1/ml] in the regions sampled
Pico PC−cyans [1/ml]
30
20
10
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

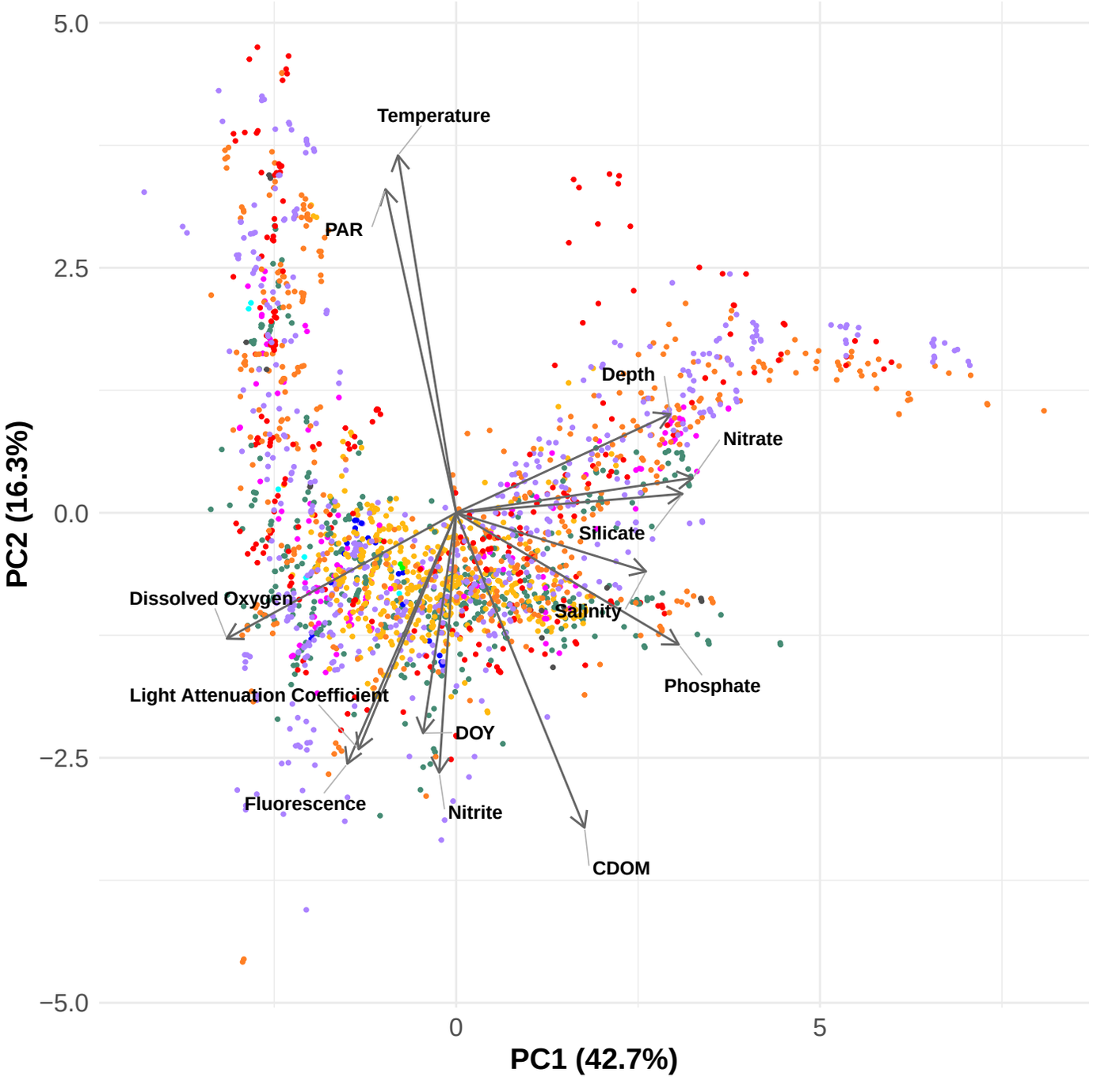

Nano PC−cyans [1/ml] in the regions sampled
Nano PC−cyans [1/ml]
60
40
20
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut


Chl maximum depths in the regions sampled as inferred from fluorescence
(condition for fluorescence: max ≥ 2 × mean)
Depth [m]
80
60
40
20
0
The North Western Passages
Gulf of Boothia
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait


Relationship between Nitrate CTD [µM] and Nitrate [µM] by region
R² = 0.92573
F = 28157
P = 0
Nitrate [µM]
Nitrate CTD [µM]
0
5
10
15
20
25
Regions
The North Western Passages
Gulf of Boothia
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
Nunavut


PCA plot of CTD and nutrient parameters for the regions sampled
Temperature
PAR
Depth
Nitrate
Silicate
Salinity
Phosphate
DOY
CDOM
Nitrite
Fluorescence
Light Attenuation Coefficient
Dissolved Oxygen
PC2 (16.3%)
PC1 (42.7%)
1.0
0.5
0.0
−0.5
−1.0
−3
−2
−1
0
1
2
Regions
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

PCA plot of significant CTD and nutrient parameters for the regions sampled



Regions

- The North Western Passages
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